Overview

Snorkel

Riga STEM Guidelines

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How to Get Started

1. Navigate to the Expert Platform
2. Once signed in, go to **My Projects**
3. Search for **"Riga,"** and locate the following projects:
    a. **EK_Riga_Production / Riga Submission**

b. **EK_Riga_Production / Riga Review** 4. Click the blue “Start” button to begin!



Submission

Riga Submission Guidelines

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Overview

Welcome to the **Riga STEM Project!** You'll be helping us develop a **dataset of new, model-stumping STEM problems** across **Physics, Chemistry, and Biology.**

Task Description

Each annotation task consists of creating an original, text-only (LaTeX-formatted) STEM problem which is run through **frontier AI models to determine difficulty** and contains the following components:

- **Problem**: A clear, single, self-contained text-only problem **written in LaTeX** that contains a STEM question that can be used to evaluate the reasoning capabilities of models. The problems must have an open solution path but a closed answer space.
    - **Note that you should not be asking multiple independent questions in your problem and linking them together. A problem is ONE question!**
- **Domain Classification and Explanation**: Your problem must fit into one of the **3 fields (Physics, Biology, Chemistry) with 15+ domains within each** and include a **justification** as to **why** it is the most appropriate selection
- **Step-by-Step Solution:** A step-by-step worked solution (also written in LaTeX) to the problem written by experts. The final answer (with units, if applicable) should appear in a LaTeX boxed environment in the final step
- **Answer:** The final correct answer that answers the problem
    - **Note**: if the problem does not specify units, the answer needs to and should include units!
- **Problem Type, Answer Type, and Concepts:** Select a description from a list of possible types that best describes the problem and answer.

Available Domains

You can choose between a **Biology, Physics, or Chemistry** problem with a corresponding subdomain on this project. Each problem should be assigned to the Domain and Sub-Domain that **best describe** the problem.

These domains should reflect the central subject matter of the **problem**, not just the notation used, the final answer format, or techniques that may appear in one possible solution.

See the Domain Description tab for further details

Quality Requirements

To be considered a high-quality submission eligible for Acceptance, the prompt must meet **all** of the following:

- **Original:** All components must be authored by human experts (LLM generated content is permitted), and all components within each sample must be original ● **Appropriate STEM Domains:** Use STEM concepts in the allowed domains ● **Challenging:** Stumps **at least one** frontier AI model majority of the time.
- **Requires reasoning:** The problem must require step-by-step reasoning to answer, rather than reciting memorised facts, theorems, etc. Solutions should be **at least 3 steps**
- **Well-formed:** All components must be grammatically or syntactically correct, and without spelling or LaTeX formatting errors
    - Currently chemfig is not supported. We plan to add support soon.
    - mhchem **is now supported on the Snorkel platform!**
- **Unique answer exists:** The problem must only have a single unique solution, but it may have multiple equivalent valid forms
    - **"Does not exist" or DNE is not a valid answer on this project**
- **Clear answer format:** The prompt should explicitly state the specified format of the answer. If the answer is meant to be parametric or expressed in terms of other variables, that should be clearly stated.
- **Reliably verifiable:** Answers to the problems and Solution Steps MUST be verifiable
- **Self-contained:** Aside from the expected STEM knowledge, problems must not contain references and must not depend on results from a single paper. Any information from tables or any background from references required to answer the question should be included in the question itself.

- **No shortcutting:** The problem must be free of hints or shortcuts that make it “trivial” to answer
- **No guessable answers:** The problem must not be easily guessable (e.g. yes/no or multiple choice)
- **Problem makes sense:** The problem itself must make sense in the context of the selected domain
- **Objectively Verifiable:** Problems should not rely on results from a single paper ● **Note**: Word puzzles are **not allowed**.

Answer Requirements

To be considered a high-quality submission eligible for Acceptance, the **answer must be correct.**

Additionally, **the answer cannot use any notation that hasn't been introduced in the prompt**. Otherwise, this violates the self-contained requirement.

Solution Requirements

To be considered a high-quality submission eligible for Acceptance, the **step-by-step solution** must meet the the following requirements:

- **Correctness:** All components must be valid and arrive at the final answer.
- **Valid Individual Steps:** Verify each solution step independently of the answer and recompute key displayed equalities and numeric value yourself. A correct final answer does not certify the steps!
- **All Claims are Supported:** For every nontrivial claim (especially after 'clearly', 'note that', 'it follows') confirm the argument is present.
    - **For example:** Check that cited theorems actually apply, check that the hypotheses are based on relevant data and sound experimental design, check for physiological relevance and plausible pathways and mechanisms, check for kinetic and thermodynamic feasibility, account for all potential intermediate reactions along with stoichiometric balance, etc.
- **All Assumptions are Explicitly Stated:** Every assumption that would change the answer to the problem if changed must be explicitly stated and justified why it is appropriate in the context of this problem.
- **Non-Computational:**
    - The problem must be solved completely without the usage of computer programming or a computer algebra system.
    - The solution must not contain excessively repetitive operations (e.g. that would be trivial with computer-assistance) or unreasonably precise computations (e.g., with many significant figures)
    - Any numerical calculations must be doable with a standard scientific calculator.
- **Accurate and Consistent Formatting:** All components must be grammatically or syntactically correct, and without spelling or LaTeX formatting errors. All expressions, variables, equations, etc. are formatted consistently across all steps of the solution.

- See Appendix LaTeX formatting for more details ● **Project Conventions:**
- Each step must start with "`Step #`"
- The last step must end with "`Final Answer: \$boxed{...}`"
- The answer in the "Answer" Box must match the exact final answer in the last step but only wrapped in $...$. The Answer box field must not include "\boxed" ○ **For example:**
  - In your last solution step: Final Answer: $\boxed{AUC}$
  - In the Answer box: AUC

Unsure on how to format the final answer in your last solution step and in the "Answer" box? Check out the Answer Formatting Guide!

Pro Tips for High-Quality Samples

- Ensure you are aligned with the above formatting requirements.
- **Double-check** your questions are distinct from one another and are not from existing datasets/competitions.
- Use good grammar, check your spelling, etc.
- Ensure your **Solution** does not include steps **within a step**, or any **trivial micro-steps**.
  - These should be broken up as much as possible, and should only cover the most **significant or substantial steps** made to reach the correct solution. You do not have to include steps that were not important in reaching the solution or points that aren't necessary in breaking down these substantial steps.
  - Defer to your best judgement here as to what should/should not be included. You can refer to the **good example steps** in the Example Problem tab as a reference for how these should be written.

LaTeX Help

Having issues with LaTeX formatting? Here are a ton of helpful resources you can utilize:

- **Overleaf:** offers **command completion** as well as a side by side source and **preview** - a great resource to reference whilst working through formatting issues
- **Codecogs:** for those **unfamiliar with** LaTeX syntax, this is an equation editor where you can click on the symbols you wish to write, and see the LaTeX code in a window, and the equation formatted in another. Please note that you will need to copy that LaTeX code and enclose it within $...$ or $$...$$ in your prompt or solution!
- **Detextify:** for those **just beginning to learn** LaTeX syntax, here you can hand draw symbols which are then returned with the LaTeX code for those symbols
- **Dainiak:** a **quick** LaTeX style **checker**
- **TeX Studio:** integrated writing environment **for creating LaTeX documents** with features including syntax-highlighting, integrated viewer, reference checking, etc.

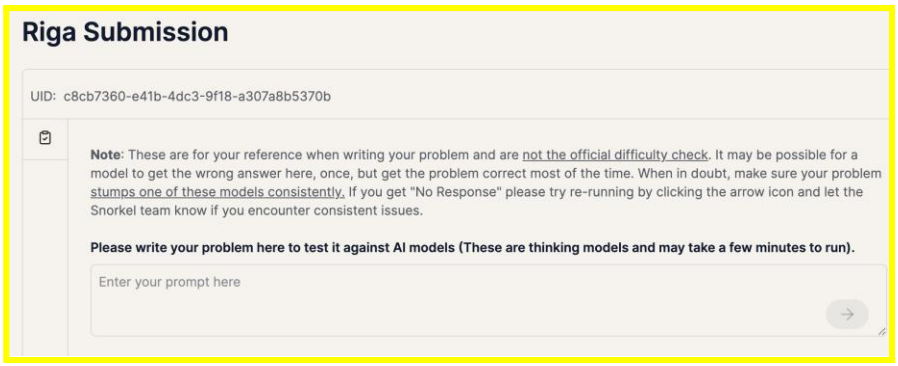
We also have four helpful youtube videos introducing LaTeX that could be helpful as well:

- **What is LaTeX?**
- **Your first LaTeX document**
- **Basic mathematics in LaTeX**
- **Intermediate mathematics in LaTeX**

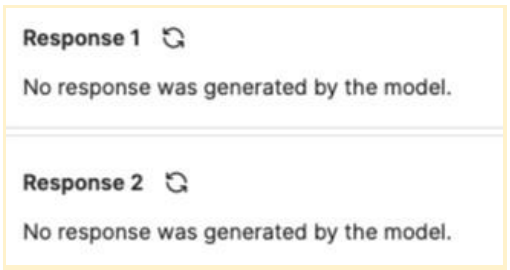
Detailed Submission Instructions

To Start

- In a **Riga Submission:**
  - You will start by testing your STEM problem against two **frontier-level "hard" models** that are intentionally difficult to stump. These models may take a few minutes to generate a response.
  - **Click the blue arrow** to run your question through the model and **ensure at least one model consistently gets the wrong answer**.
  - Enter your STEM problem in LaTeX in the text box that states: *"Enter your prompt here."*



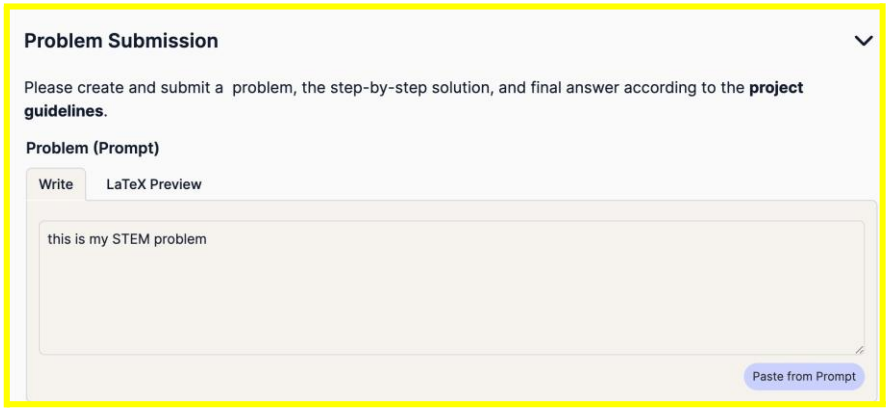
⚠PLEASE NOTE: "No Response was generated by the model" is not equivalent to the model getting an incorrect answer. If you encounter this, please re-run the models.



Submitting the Question

Once you're confident you've created a problem that **consistently stumps one of the provided models**, you'll complete the rest of the task on the right side of the screen.

- Use the copy and paste your prompt into the text box under the **"Problem (Prompt)"** heading on the right-hand side of the UI.



- Select the appropriate **domain and sub-domain** for the question

- Fill out the **Domain Explanation field** by providing a brief and clear explanation of why this domain and subdomain are the best choice for your STEM problem and why it's more appropriate than the next best domain and subdomain. Justifications should follow the below structure:

  This problem involves <description of the key elements of the problem>, which are part of <domain and sub domain>.

  The problem also involves <description of the additional elements of the problem>,

  which are part of <other domain and/or sub domain>.

  However, <explanation of why they're not core to the problem or less advanced>.

Worked Solution

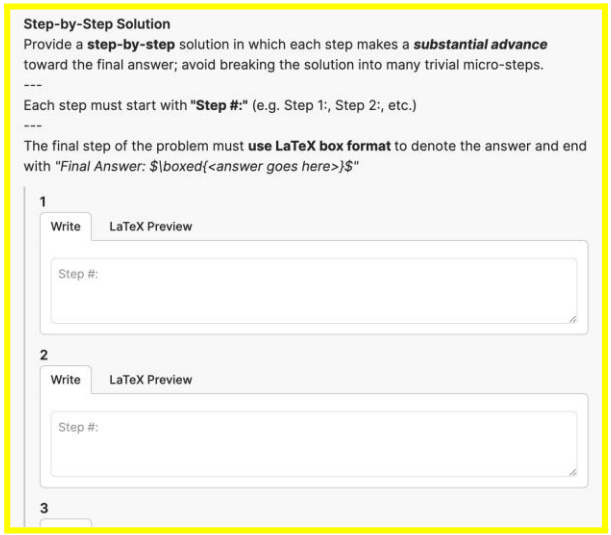
- Provide a step-by-step solution in which each step makes a substantial advance toward the final answer; avoid breaking the solution into many trivial micro-steps.

The final step of the problem must use LaTeX box format to denote the answer and end with **"Final Answer: $\boxed{<answer goes here>}$"**

- Ensure that each step of your solution can be verified by the Reviewer **independently of the answer** - recompute key displayed equalities and numeric values yourself prior to submitting. A **correct final answer does not certify the steps!**
  - For every nontrivial claim (especially after 'clearly', 'note that', 'it follows') **confirm your argument is present**. Check that cited theorems actually apply.
  - Every assumption that would change the answer to the problem if changed must be **explicitly stated and justified** why it is appropriate in the context of this problem.

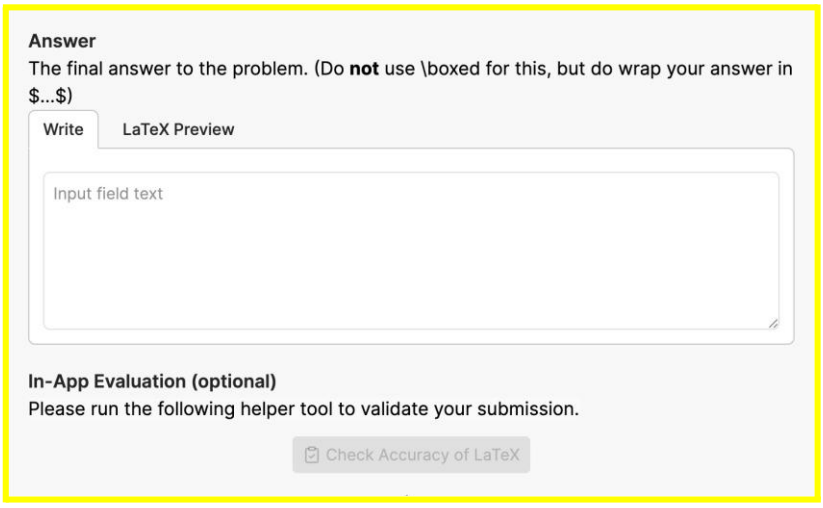
- Provide **all necessary reasoning steps to advance the solution** in the **"Worked Solution"** field.
  - There should be a minimum of at least **3** steps.
  - Each **step** must make a significant advancement toward reaching the final answer rather than simply breaking down the solution into small/trivial pieces.
  - Do not include steps that were not inherently important or points that are not necessary to understanding how the solution was reached.
  - Ensure each Step is just that, **one step**. Do not include **steps within a step, substeps, micro-trivial steps, or small/trivial content** that isn't an **absolutely important** or necessary step in reaching the final answer.

  - You **must** enumerate each worked solution step using the following format: *Step 1, Step 2, Step 3,* etc
  - All worked solution steps must be **explicitly numbered in sequence** (*Step 1, Step 2*, …)
  - If you are referencing a **previous step** (i.e., discussing something from Step 2 that is being recalled or referenced in Step 4, you should explicitly mention that as "Step 2")



Answer Section

- Add in **only** the final answer (with units, if applicable) to the problem (and do **NOT** use \boxed in this field!)
- Once done, run the In-App Evaluation button to validate your submission
    - Please disregard the “(optional)” portion of the header; validating your submission is **mandatory for this project**!



- See the Example Problem tab for good examples to reference!

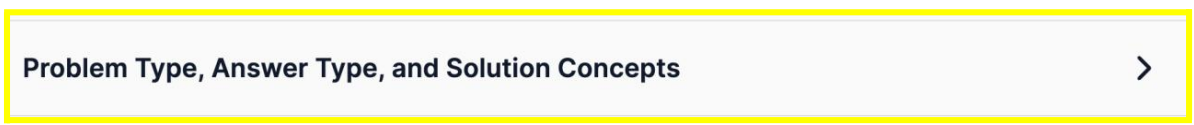
Problem, Answer, and Concepts Section

- Lastly, you will select the most appropriate problem and answer classification. Select the options that best describe your problem based on the Answer Types and Problem Types descriptions above!

- You will also be asked to list out the **concepts, techniques, or knowledge** required to solve your problem. This must include **at least one and no more than five**. If more than five apply, you’ll be asked to list the **five most important.** Lists should follow the below structure:

For **example**, these concepts could be listed for a calculus problem:

- *Critical points*
- *First-derivative sign changes*
- *Local maxima/minima*
- *Endpoint comparison*
- *Polynomial differentiation*



Submission Routing

In this section, you will be able to select **where would you like your submission** to go after you submit, with the following options:

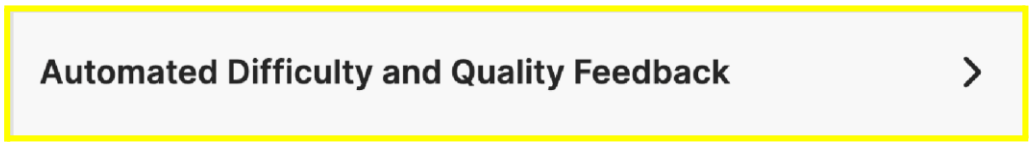
- Select **"Quality Review"** when you believe your submission **meets project quality standards**, passes automated checks, and is ready for quality review by a human reviewer
- Or, you can select the optional **"Automated Checks Only"** to check your **in-progress submission** against the automated checks for **difficulty and originality**. This will be an indicator as to whether or not your submission is ready for quality review!



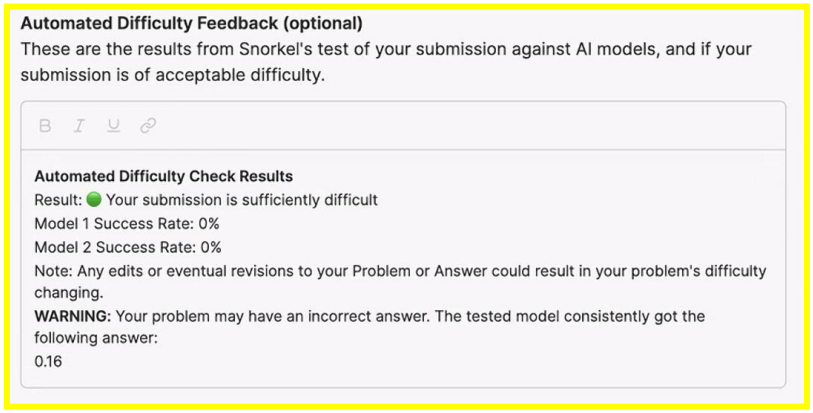
Automated Difficulty and Quality Feedback

If you do elect to send your submission through the optional **“Automated Checks Only,”** you will be able to see the results of these automated checks while still working on your submission, including:

- Whether your submission is of acceptable difficulty
- The full results of how the models solved your problems
- Whether your problem and solution are correct and complete
- Whether your solution steps are correct and if there are logical gaps or mathematical errors
- Whether your selected domain and justification are appropriate



- For example, one of the checks:



Final Checks

- **Final Checklist:** Confirm the following:
  - ✅There is **one and only one correct answer.**
  - ✅You have **checked for spelling and grammar.**
  - ✅You have **independently verified the accuracy of your solution steps.**
  - ✅The question is **not plagiarized.**
  - ✅You have used appropriate **LaTeX formatting.**
  - ✅You have ensured the Answer box field has the final answer **only** in the correct format (Do not use \boxed for this, but do wrap your answer in $...$) see Answer Formatting Guide

Submission

- When all fields are complete, click **Submit**.

Review

Riga Review Guidelines

Overview
Task Overview
Reviewer Responsibility
Quality Requirements
Tips for High Quality Reviews
Revision or Rejection Notes
Step-by-Step System Walkthrough

Overview

As a **reviewer**, your responsibility is to ensure submitted problems meet the quality standards of this project. This means checking the following:

1. Is the prompt appropriate for the project?
2. Is the answer correct?
3. Are the solution steps complete, error-free, and well-formatted?
4. Are the labels and categories reasonably accurate?

You are responsible for ensuring the answer to all of these questions is **Yes**. If it's not, you must provide **clear, actionable feedback** to the original author that helps them improve the submission. We expect reviews to take 20-30 minutes on average.

Reviewer Responsibility

Each review task involves validating a submitted problem created by an Original Author (OA).

You are **not rewriting or re-authoring content**. Your job is to ensure the Original Author's submission meets **ALL** high-quality standards, **validate accuracy/guideline adherence,** correct minor issues (grammar, formatting, spelling, clarity only), and provide your own evaluation in the **Reviewer Section** and leave feedback**.**

When leaving feedback for the OA, you are both helping them correct the specific submission and **coaching them to improve**. Reviewers are expected to follow a strict structure to ensure feedback is **thorough, actionable, and encouraging**.

ℹ Please navigate to the **Revision or Rejection Notes** below for full details on expectations for feedback.
At the end of the review you will select one of the following options:

- **Accept** - The submission meets all quality requirements and needs no further edits or reviews

- **Needs Revision -** The submission needs changes or updates by the original author before it can be considered acceptable

- **Reject -** The submission is obvious spam and completely invalid for the project
    - **Warning:** Rejections flag the submitter for review by the Snorkel team. These should be rare and only used in cases where the submissions is obviously spam, plagiarised, or inappropriate



Prompt Requirements

To be considered a high-quality submission eligible for Acceptance, check that the prompt meets **all** of the following:

- **Well-formed:** All components must be grammatically or syntactically correct, and without spelling or LaTeX formatting errors
- **Original:** All components must be authored by human experts (LLM generated content is permitted), and all components within each sample must be original
- **Unique answer exists:** The problem must only have a single unique solution, but it may have multiple equivalent valid forms
    - "Does not exist" or DNE is not a valid answer on this project
- **Clear answer format:** The prompt should explicitly state the specified format of the answer. If the answer is meant to be parametric or expressed in terms of other variables, that should be clearly stated.
- **Reliably verifiable:** Answers to the problems and Solution Steps MUST be verifiable
- **Self-contained:** Aside from the expected STEM knowledge, problems must contain all the necessary context required to answer the problem. Tables, reference material, etc. that are required for answering must be included in the problem.

- **No shortcutting:** The problem must be free of hints or shortcuts that make it “trivial” to answer
- **Requires reasoning:** The problem must require step-by-step reasoning to answer, rather than reciting memorised facts, theorems, etc.
- **No guessable answers:** The problem must not be easily guessable (e.g. yes/no or multiple choice)
- **No proofs**: Prompts that ask “show that”, “prove”, “verify”, etc. are not allowed.
- **Problem makes sense:** The problem itself must make sense in the context of the selected domain
- **Note**: Word puzzles are **not allowed**, even if solvable with set theory.

❓During your reviews, If you are unsure as to whether or not a submission **would be considered Acceptable**, please don’t hesitate to reach out in the **#ec-riga-reviews channel** - a Riga Team Member will be happy to support and talk through a review with you!

Answer Requirements

To be considered a high-quality submission eligible for Acceptance, the **answer must be correct.**

Additionally, **the answer cannot use any notation that hasn't been introduced in the prompt**. Otherwise, this violates the self-contained requirement.

ALWAYS answer the question yourself. If it is different from the OA’s, you will explain the errors in the OA’s logic when you select Needs Revision or Reject.

Solution Requirements

To be considered a high-quality submission eligible for Acceptance, the **step-by-step solution** must meet the following requirements:

- **Correctness:** All components must be mathematically/objectively valid and arrive at the final answer.
- **Valid Individual Steps:** Verify each solution step independently of the answer and recompute key displayed equalities and numeric value yourself. A correct final answer does not certify the steps!
- **All Claims are Supported:** For every nontrivial claim (especially after 'clearly', 'note that', 'it follows') confirm the argument is present. Check that cited theorems actually apply, that induction is proven (not pattern-matched), and that case splits cover all cases including boundary/sign/degenerate ones.

    - **For example:** Check that cited theorems actually apply, check that the hypotheses are based on relevant data and sound experimental design, check for physiological relevance and plausible pathways and mechanisms, check for kinetic and thermodynamic feasibility, account for all potential intermediate reactions along with stoichiometric balance, etc.
- **All Assumptions are Explicitly Stated:** Every assumption that would change the answer to the problem if changed must be explicitly stated and justified why it is appropriate in the context of this problem.
- **Accurate and Consistent Formatting:** All components must be grammatically or syntactically correct, and without spelling or LaTeX formatting errors. All expressions, variables, equations, etc. are formatted consistently across all steps of the solution. If the problem does not specify units, ensure the answer includes them.
- **Non-Computational:**
    - The problem must be solved completely without the usage of computer programming or a computer algebra system.
    - The solution must not contain excessively repetitive operations (e.g. that would be trivial with computer-assistance) or unreasonably precise computations (e.g., with many significant figures)
    - Any numerical calculations must be doable with a standard scientific calculator.
- **Project Conventions:**
    - Each step must start with "`Step #`"
    - The last step must end with "`Final Answer: \$boxed{...}`"
    - The answer in the "Answer Box" must match the final answer in the last step.

Labels and Categories

Each problem has a domain, sub-domain, problem type, and answer type. In general, as long as these are not obviously incorrect they can be accepted but please follow the field-specific guidance below:

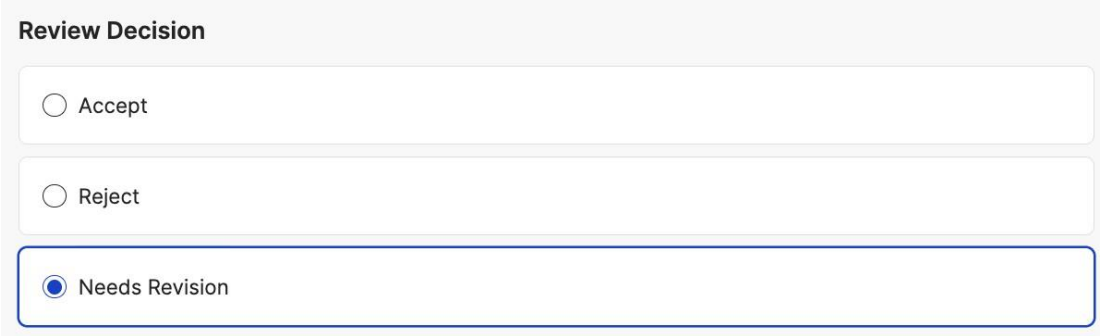
- **Domain and Sub Domain:** If the selected domain, sub-domain, and explanation are completely inaccurate, select "Needs Revision" and include comments on the issue in your feedback. If it's plausible or borderline for the problem to be considered in the selected domain and sub-domain, then you may leave as-is.
- **For Problem Type and Answer Type:** If you think another option would be more appropriate, please make the change yourself and note to the OA if you are writing "Needs Revision" feedback.
- **For Concepts List:** If you think another option would be more appropriate, please make the change yourself and note to the OA if you are writing "Needs Revision" feedback.

Tips for High-Quality Reviews

- Keep your feedback **clear, logical, and accurate**. Always provide **specific**, **actionable suggestions**!
- Important Checklist:
    - Look for a clear walk-through of the correct answer in the **Step-by-Step Solution**
        - **And ensure these steps are enumerated!**
    - Grammar and fluency in both the question and answer
- **Emphasize making light edits that help clarify answer correctness but do not change any of the original content.**
- If the OA answer is wrong, your answer should clearly explain why OA is wrong, what the correct answer is, and why your answer is correct in the **"Review Section.**"
- If in doubt about a **borderline** issue, **add a note** in the **"Review Section"** field. The team will review it.

Ask yourself: *If the Original Author addressed everything I included in the revision notes, would it be acceptable upon re-submission? If not, more needs to be added!*

Revision or Rejection Notes



Overview

- If you decide to send the submission back to **Needs Revision**, fill out the **Revision Notes** field for the Original Author to view. Reviewers **MUST** provide thorough, actionable, and encouraging feedback to help submitters revise and re-submit an acceptable submission.
- If you decide to Reject the submission, fill out the **Rejection Notes** field for the Original Author to view. Reviewers **MUST** provide thorough, actionable, and encouraging feedback to help submitters learn from their mistakes and improve upon a new submission.

New Feedback Workflow

- As noted in the **Review Process Updates** tab, we have implemented **Error Categories** into your review tasks for you to specifically categorize the errors you identify. You will

see an option to select these Error Categories once you have selected **“Needs Revision” or “Reject”** in the **Review Decision** dropdown

- Once Error Categories have been selected, these should be **explicitly mentioned in your feedback**, with **specific suggestions on how to revise each**. Review Feedback that is generic, vague, or unhelpful and does not provide actionable suggestions for each Error Category marked will be considered a **low-quality review**.

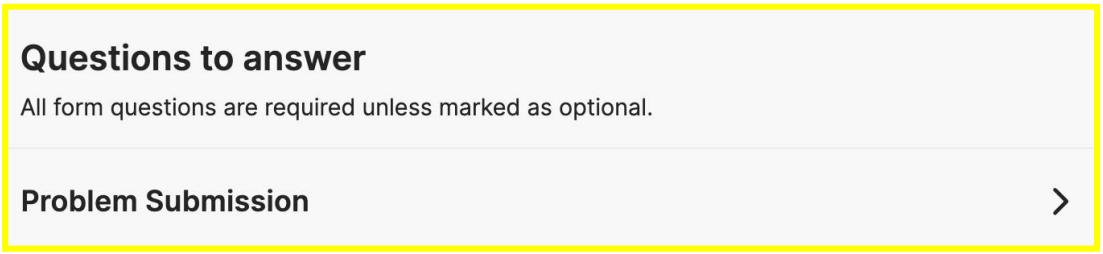
Step-by-Step System Walkthrough for Reviews

Step 1: Click “Start” on Riga Review



Step 2: Review the Problem, Answer, and Solution

Upon opening a Riga Review, start by opening the **“Problem Submission”** section. This will include the problem, domain, step-by-step-solution, and answer. **This is where the majority of your time will be spent reviewing.**



Step 3: Review the Problem & Answer Types and Concept List

This section contains the categories from the problem/answer and concepts list.

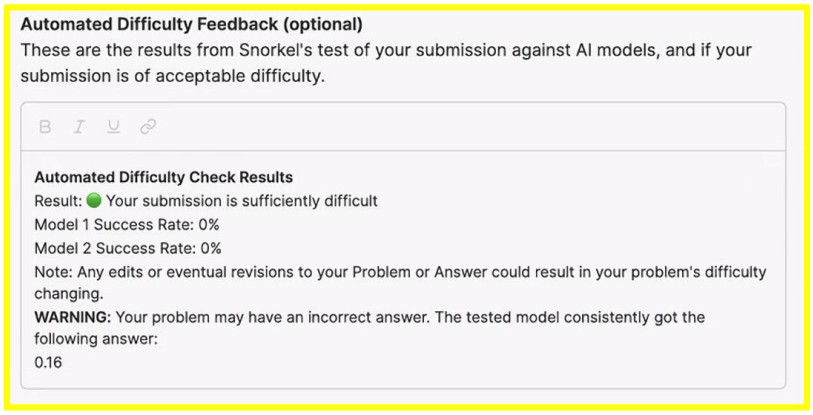


Step 4: Gather More Context from the Automated Checks

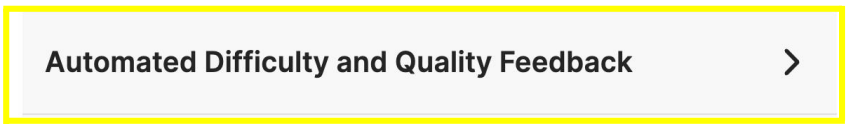
We have added a new section titled **“Automated Difficulty and Quality Feedback”** that will allow you to **read through Snorkel’s automated checks** the submission was run through to

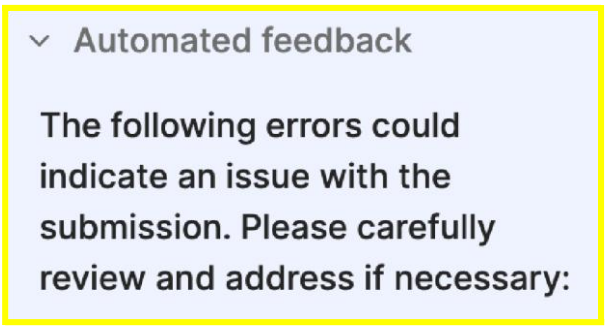
give you more context as to how each aspect of the submission performed. These checks include:

- Whether the submission is of acceptable difficulty
    - **Note**: Reviewers **DO NOT** need to assess the difficulty of a task. Assume that if a task makes it to human review, the problem has been **deemed sufficiently difficult** enough by our automated checks, and you do not need to assess any further.
    - **HOWEVER**: if you notice that a problem is **blatantly trivial**, or is clearly using tricks or specific setups/strategies to stump the models (**for example**: a very basic Chemistry concept that is setup as a word puzzle that stumps the model based on an ambiguous setup, not by the model actually attempting reasoning and producing the wrong answer) it is valid to send back for revision. If you are unsure, please reach out in the **#ec-riga-reviews** channel to discuss!
- The full results of how the models solved the problem
- Whether the problem and solution are correct and complete
- Whether the solution steps are correct and if there are logical gaps or mathematical errors
- Whether the selected domain and justification are appropriate ● For example, one of the checks:



Read through all of the automated checks to gain insight into potential issues called out by our internal evaluations. Once you have read through **all of the submission content**, the automated checks, and attempted to verify the solution and answer, **move on to Step 5 Additionally:** you will be able to see a blue **“Automated Feedback”** dropdown on the left-hand side of your review task. This will include additional automated checks to provide context as to what issues the submission could potentially have. Please use both the **“Automated Feedback”** field **AND** the **“Automated Difficulty and Quality Feedback”** section for insight while completing your assessment.



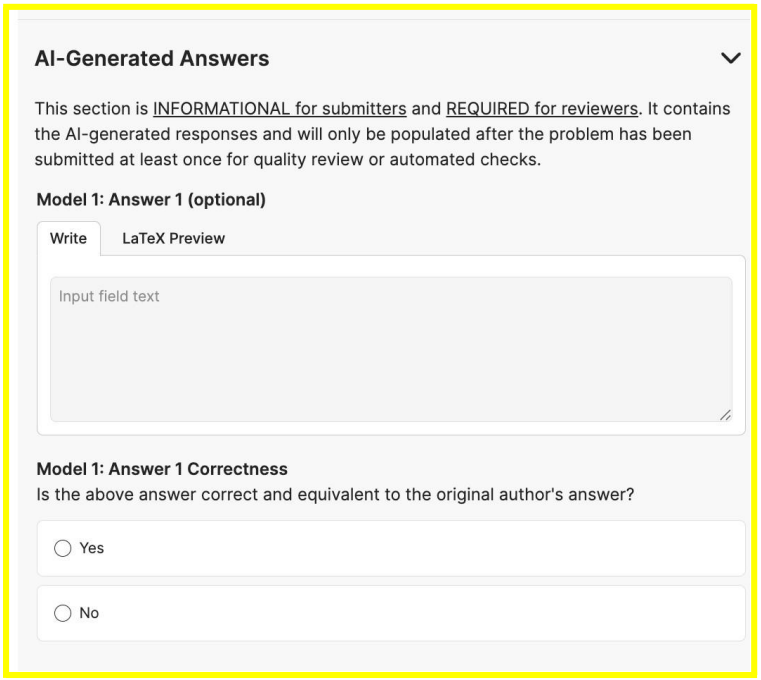


Step 5: Make Minor Edits as Necessary

Once you have read through and fully understood the content of the submission, prior to attempting to verify/answer the question yourself, make **light grammar, formatting (LaTeX), spelling, and clarity edits** to the Original Author's solution steps, **if needed** (Reviewers will not revise the problem or answer).

Step 6: Review the Correctness of AI Generated Answers

In this **mandatory section**, you will be asked to review the model-generated answers for accuracy and confirm whether they are correct or incorrect.



Step 7 : Fill out the Review Section



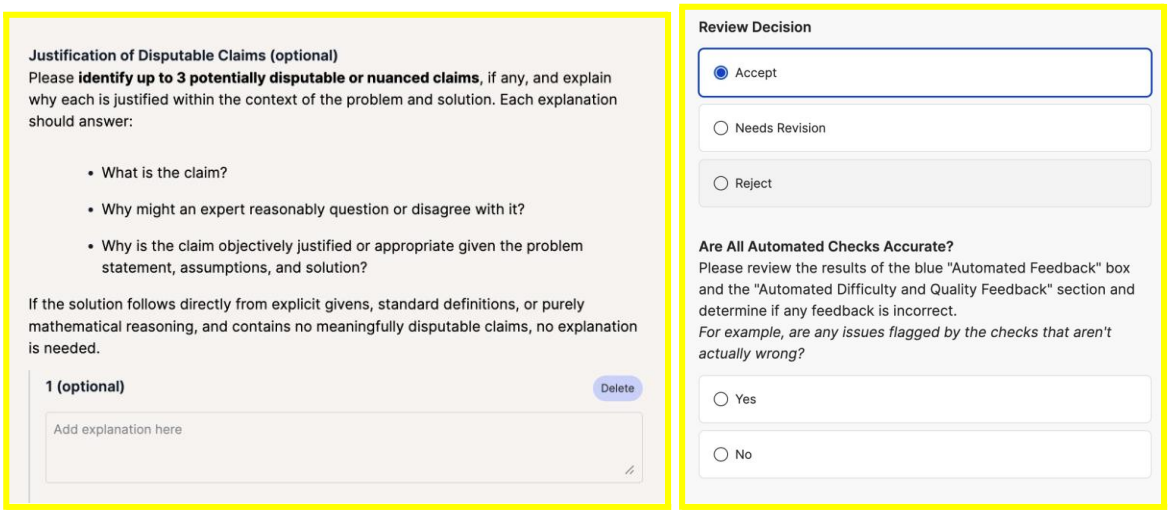
This is where your evaluation begins. You will be asked to indicate whether you **Accept** the submission, want to send it back for **Needs Revision**, or **Reject**.

If **Accept**:

- Ensure that the problem, solution and answer **meet all stated criteria in this document** (i.e., appropriate level, self-contained, reasoning not recall, single solution, not a proof, not excessively computational, not dependent on obscure theoretical knowledge, etc)
- The answer and step-by-step solution are certifiably correct, and are **verified independently from one another (remember: a correct answer does NOT verify the solution steps!)**
- The problem aligns with the selected domain and subdomain, with a well-written justification that leaves no doubt
- Some problems and solutions may include claims, assumptions, or interpretations that a domain expert could reasonably question. For each problem, identify up to 3 potentially disputable or nuanced claims, if any, and explain why each is justified within the context of the problem and solution. For each claim, answer: ○ What is the claim?
    - ○ Why might an expert reasonably question or disagree with it?
    - ○ Why is the claim objectively justified or appropriate given the problem statement, assumptions, and solution?

  If the solution follows directly from explicit givens, standard definitions, or purely mathematical reasoning, and contains no meaningfully disputable claims, no explanation is needed.

- Upon clicking Accept, Reviewers will be prompted with an additional question: **Are All Automated Checks Accurate?**
    - ○ This will ask you to indicate whether the results of the blue **“Automated Feedback”** box and the **“Automated Difficulty and Quality Feedback”** section mentioned in Step 4 are all accurate. If you deem the checks are accurate based on your analysis, select **“Yes,”** and submit the review task. If you select **“No”:**
    - ○ You’ll see one final, additional ask that will request an explanation on **why the submission is acceptable despite the automated check inaccuracies**. Please provide a justification as to why the potential inaccuracies (or false positives) **do not affect the submission enough** to keep it from being accepted.
- Once done, submit the review task.



If **Needs Revision**:

- If you identify any errors that **could not be revised with minor edits** (edits exceeding minor being any that would require the rewriting/re-authoring/re-structuring of the submission content so that it changes the meaning or fundamentally alters the problem/solution), opt to send the task back to Needs Revision
- Once selected, Reviewers will be prompted with **selecting all Error Categories that apply**. This will be a comprehensive list that allows you to label which errors you came across in your assessment.
    - See the **Error Category Definitions** tab for full details to assist you in making your selections
- When all Error Categories have been selected, leave **Revision Notes** for the OA. Read over the following Review Guidelines prior to writing your feedback:

    - **Revision or Rejection Notes**

- Revision Notes **MUST** include **explicit mention of the Error Categories selected**, and provide specific, actionable suggestions for the OA to work off of. See the following example of **poor Reviewer Feedback:**

"Your problem is not self-contained. Step 3 relies on repeated 'The Euclidean reductions gives,' which is not self-contained and thus not reproducible."

    - This feedback provides no specific suggestions, is discouraging, and would not give the OA any direction **as to how to get their submission accepted.**
    - Reviewers should ask themselves: **If the Original Author addressed everything I included in the revision notes, would it be acceptable upon re-submission?** If **not**, more needs to be added!
- Once done, submit the review task.

If **Reject**:

- Ensure you are only rejecting for clear **spam, low effort, or completely ignoring project rules.**
- Provide specific, thorough, and encouraging feedback to the OA that will motivate them to improve their next submission.
- Rejection Feedback like: *“Clear spam, the task must be rejected”* would be considered low-quality feedback. Ensure you are identifying the **specific reason for rejection** and why it warrants an outright rejection as opposed to Needs Revision.
- Once done, submit the review task.

Review Process Updates

Review Process Updates: PLEASE READ

For those who have worked on previous Snorkel STEM projects, the review process has undergone several changes aimed at improving and clarifying this workflow so **all expectations are explicitly clear** and reviews are of the **highest possible quality**. Please read through the following updates:

#1. Double-Down on Solution Steps:

The solution steps are often the largest reason for a task not meeting the Acceptance bar, and being sent back for revision. Moving forward, **Reviewers must do the following:**

- Verify each step independently of the answer. A **correct final answer does not certify the steps!**
- For every **nontrivial claim** (especially after 'clearly', 'note that', 'it follows') **confirm the argument is present**.

#2. Selecting the New "Error Categories":

To improve the consistency of review feedback, we have implemented **new Error Categories** in your review tasks that will allow you to **select specific labels for errors you identify**. Reviewers will be required to select these categories and ensure their revision feedback corresponds with what was selected.

- See **Error Category Definitions**
- Reviewers also **MUST** provide specific, actionable suggestions to revise any identified error categories. Feedback like *"make the problem more self-contained"* will **not be tolerated** on this project.
    - If a problem were found to not be self-contained, a Reviewer should select the **"Not Self-Contained"** error category, mention it **explicitly** in the revision feedback, include actionable details regarding **what makes the current question not self-contained** and provide **specific suggestions**/additional information as to what should be added to the question background to resolve the problem!
- Every time a Reviewer leaves feedback, the question should be: **If the Original Author addressed everything I included in the revision notes, would it be acceptable upon re-submission?** If not, **more** needs to be added!

#3. Justifying Key Claims and Reasoning Steps

Some problems and solutions may include claims, assumptions, or interpretations that a domain expert could reasonably question. For each problem, **identify up to 3 potentially disputable or nuanced claims**, if any, and explain why each is justified within the context of the problem and solution.

For each claim, answer:

- What is the claim?
- Why might an expert reasonably question or disagree with it?
- Why is the claim objectively justified or appropriate given the problem statement, assumptions, and solution?

If the solution follows directly from explicit givens, standard definitions, or purely mathematical reasoning, and contains no meaningfully disputable claims, no explanation is needed.

#4. Transparent Reviewer Performance Feedback

In Riga, Reviewers will be provided a transparent performance feedback system that will allow for continued improvement and closer collaboration with our team to ensure alignment on expectations. This will specifically include **Weekly "Summaries"** to Reviewers on review performance, including: errors that were missed, areas of misalignment, areas to improve on, areas of strength, and a full performance breakdown. We will track overall performance, which will be grouped into the following statuses:

- **Trial Period**
  - Once a Reviewer passes the Reviewer Assessment, they will begin with a **"Trial Period" of 3-5 days**. This will include a **throttled daily review limit** of **25 per day** to allow for a smooth transition into reviewing as you adjust to the guidelines/workflow. ● **Unlimited (No Throttle)**
  - After the Trial Period, once a Reviewer has consistently produced high-quality reviews and demonstrated a clear, comprehensive understanding of the review workflow and expectations, they will be given **"Unlimited"** status. Reviewers will be able to complete the maximum amount of daily reviews, **40**. ● **At Risk (Throttled)**
- With the Weekly Summaries provided, Reviewers will also be able to see the **% of problems they reviewed that were found to have had errors** compared to their total review count. If a Reviewer reaches more than **10% of errors** across all reviews (which includes **ANY** errors that slipped through, including minor issues like LaTeX) this will warrant a **violation**, and they will be placed in an **"At Risk"** status.
- This violation will include a **significant decrease of the Reviewer's daily limit**, as well as an individual coaching message with specific examples of the errors identified to assist the Reviewer in improving. This status will be lifted once the Reviewer demonstrates consistent improvement on the previously identified errors.
- If a Reviewer's error percentage **continues to increase during "At Risk" status**, and improvement is not shown, this will potentially result in a **removal of review access in Riga**.

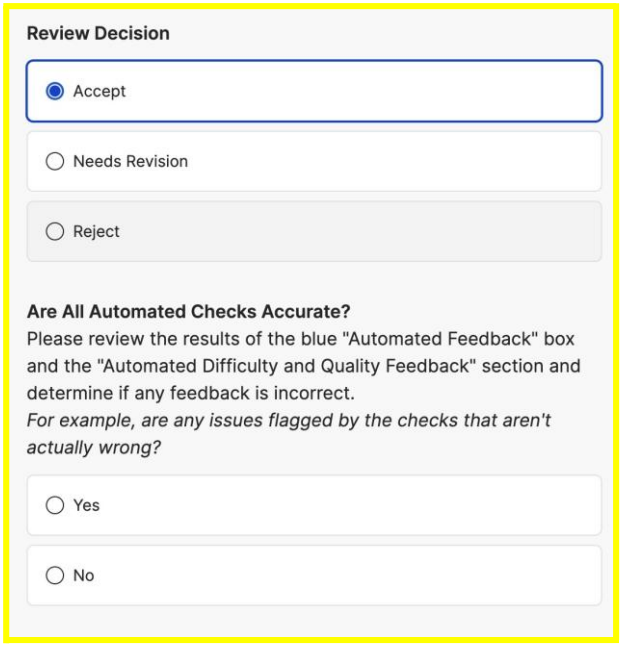
Note: a violation can also be warranted if revision feedback is found to be generic, vague, unhelpful, or misaligned with the expectations outlined here

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#5. Mandatory Automated Feedback Check

Reviewers will now be required to look over the complete automated feedback checks on a submission **in a review task**. The purpose of this is to provide reviewers more information/context on the submission itself, as well as how our automated checks work to help **improve your understanding** of each aspect of the review workflow. Doing these checks will allow reviewers to be more comfortable working with evals:

- Upon selecting "**Accept**" for the Review Decision, Reviewers will be prompted with answering another question: **Are All Automated Checks Accurate?**
- Reviewers will note whether any of the automated checks are **inaccurate**, including false positives, but indicate why the task can **still be Accepted**
- See more details in the **Step-by-Step System Walkthrough** tab!



Error Categories

Error Category Definitions

When a submission is flagged as “Needs Revision” you must identify one or more error categories that are driving the review.

Submission Area	Error Category	Definition
Prompt	Not Self-Contained	Prompt is missing definitions, formulas, or context needed to attempt the problem
	Non-Unique Answer	As written, the prompt would yield multiple valid answers
	Too Computational	Problem would require code or machine computation to verify
	Multi-Part Question	Prompt asks multiple questions or multiple parts of a single question
	Guessable	Prompt is multiple-choice, true-false, or setup in such a way that the answer can be easily guessed
	Domain Beyond Project Scope	The prompt is a STEM domain outside of what is allowed for this project
Answer and Solution Steps	Wrong Answer	Wrong answer or flawed, inaccurate reasoning
	Unjustified or Unsupported Claims	Solution skips steps, assertions are made without justification, etc. such that the work shown cannot be fully understood or reproduced
	Unstated Assumption	Every assumption that would change the answer to the problem if changed must be explicitly stated and justified why it is appropriate in the context of this problem.
	Step Error	There is an error in an individual step but the final answer is still correct

All	Major LaTeX or Formatting Issues	Significant use of wrong delimiters, unsupported commands, missing project-specific formatting, etc. ***NOTE:*** *Minor LaTeX or grammar errors*
		*and/or small typos in the solution step should be fixed in review and, by themselves, are not grounds for flagging a submission for needs revision. If you are sending a submission back for needs revision for **other** reasons and did make minor LaTeX or grammar revisions, ensure you leave a note explaining the changes made.*
	Major Grammar or Language Issues	Consistently unclear phrasing, awkward sentence structure, spelling errors, and/or imprecise mathematical language that makes the intent ambiguous or difficult to follow ***NOTE:*** *Minor LaTeX or grammar errors and/or small typos in the solution step should be fixed in review and, by themselves, are not grounds for flagging a submission for needs revision. If you are sending a submission back for needs revision for **other** reasons and did make minor LaTeX or grammar revisions, ensure you leave a note explaining the changes made.*
	Other	Wrong domain, closed quota, missing justification, any other errors identified not included in this list

⚠**Note**: you should only **Reject** a submission for **clear spam, low effort, or completely ignoring project rules**.

Problem and Answer Reference

Problem Types

[Biology](#)
[Chemistry](#)
[Physics](#)

(jump to [Answer Types](#))

Biology: Problem Form Taxonomy

Problem forms describe the primary type of answer a problem asks for, independent of biological domain or subdomain. Each problem should be assigned the form that best reflects what the final response must provide and how that response can be verified. These forms should avoid proof, derivation, or essay-style responses; expected answers should be directly checkable.

- **Exact computation** - A problem asking for an exact numerical value, count, ratio, probability, genotype frequency, allele frequency, sequence length, cell count, population size, concentration, rate, percentage, coefficient, or other exact biological quantity derived from the given information.
- **Symbolic expression** - A problem asking for a closed-form expression, biological relationship, growth model, inheritance formula, population model, rate expression, regulatory relationship, or symbolic representation, without requiring the derivation steps.
- **Numerical approximation** - A problem asking for a numerical answer computed to a specified tolerance, precision, number of significant figures, number of decimal places, or error bound.
- **Solve for unknowns** - A problem asking for one or more values, genotypes, phenotypes, rates, frequencies, concentrations, parameters, species, taxa, sequences, or biological states that satisfy a given condition, dataset, model, or biological relationship.
- **Construction under constraints** - A problem asking for any valid biological object or representation that satisfies a stated set of genetic, molecular, ecological, anatomical, experimental, or quantitative constraints.
- **Optimization** - A problem asking for an optimal value, condition, trait, allocation, sequence, experimental parameter, growth condition, threshold, fitness value, or biological outcome under a specified objective and constraint set.
- **Exhaustive enumeration** - A problem asking for the complete set, list, count, or classification of all valid genotypes, phenotypes, taxa, species, sequences, codons, mutations, cell types, pathway states, or experimental outcomes satisfying the given criteria.
- **Canonicalization or normalization** - A problem asking for a biological object to be rewritten in a specified standard, reduced, normalized, or canonical form, such as a DNA sequence orientation, RNA transcript, amino acid sequence, genotype notation, taxonomic name, pathway notation, or normalized dataset value.

- **Transformation between representations** - A problem asking for conversion of an object from one biologically equivalent representation to another, such as DNA to mRNA, mRNA to protein, genotype to phenotype, pedigree to inheritance mode, sequence to alignment, tree to relationship statement, or raw counts to normalized expression.
- **Parameter identification** - A problem asking for unknown parameters, constants, rates, coefficients, fitness values, allele frequencies, growth rates, binding constants, kinetic parameters, model terms, or latent biological quantities to be inferred from equations, data, constraints, or observed biological behavior.

Chemistry: Problem Form Taxonomy

Problem forms describe the primary type of answer a problem asks for, independent of chemical domain or subdomain. Each problem should be assigned the form that best reflects what the final response must provide and how that response can be verified. These forms should avoid proof, derivation, or essay-style responses; expected answers should be directly checkable.

- **Exact computation** - A problem asking for an exact numerical value, formula value, stoichiometric quantity, concentration, amount, yield, charge, energy, equilibrium constant, rate quantity, coefficient, or other exact chemical object derived from the given information.
- **Symbolic expression** - A problem asking for a closed-form expression, chemical relationship, rate law, equilibrium expression, thermodynamic expression, formula, or symbolic representation, without requiring the derivation steps.
- **Numerical approximation** - A problem asking for a numerical answer computed to a specified tolerance, precision, number of significant figures, number of decimal places, or error bound.
- **Solve for unknowns** - A problem asking for one or more values, concentrations, pressures, temperatures, amounts, coefficients, species, products, structures, or variables that satisfy a given equation, system, reaction condition, constraint, or chemical relationship.
- **Construction under constraints** - A problem asking for any valid chemical object that satisfies a stated set of structural, compositional, stoichiometric, thermodynamic, spectroscopic, or quantitative constraints.
- **Optimization** - A problem asking for an optimal value, condition, composition, reagent amount, separation condition, yield, pH, temperature, pressure, or experimental parameter under a specified objective and constraint set.
- **Exhaustive enumeration** - A problem asking for the complete set, list, count, or classification of all valid products, species, isomers, resonance forms, configurations, mechanisms, phases, or parameter values satisfying the given criteria.
- **Canonicalization or normalization** - A problem asking for a chemical object to be rewritten in a specified standard, reduced, normalized, or canonical form, such as a balanced equation, IUPAC name, empirical formula, net ionic equation, oxidation-state assignment, or canonical SMILES string.

- **Transformation between representations** - A problem asking for conversion of an object from one chemically equivalent representation to another, such as name to formula, formula to structure, structure to SMILES, molecular equation to net ionic equation, spectrum to structure, or composition data to empirical formula.

Parameter identification - A problem asking for unknown parameters, constants, coefficients, rate constants, equilibrium constants, thermodynamic quantities, analytical calibration terms, structural values, or latent quantities to be inferred from equations, data, constraints, or observed chemical behavior.

Physics: Problem Form Taxonomy

Problem forms describe the primary type of answer a problem asks for, independent of physical domain or subdomain. Each problem should be assigned the form that best reflects what the final response must provide and how that response can be verified. These forms should avoid proof, derivation, or essay-style responses; expected answers should be directly checkable.

- **Exact computation** - A problem asking for an exact numerical value, formula value, vector, field value, energy, force, momentum, frequency, wavelength, coefficient, probability, or other exact physical quantity derived from the given information.
- **Symbolic expression** - A problem asking for a closed-form expression, physical relationship, equation of motion, field expression, scaling law, transfer function, rate expression, or symbolic representation, without requiring the derivation steps.
- **Numerical approximation** - A problem asking for a numerical answer computed to a specified tolerance, precision, number of significant figures, number of decimal places, or error bound.
- **Solve for unknowns** - A problem asking for one or more values, vectors, fields, times, positions, velocities, energies, frequencies, parameters, states, or physical quantities that satisfy a given equation, system, constraint, model, or physical condition.
- **Construction under constraints** - A problem asking for any valid physical configuration, state, circuit, trajectory, field, wave, model, parameter set, or experimental setup that satisfies a stated set of physical or quantitative constraints.
- **Optimization** - A problem asking for an optimal value, condition, trajectory, design parameter, efficiency, frequency, angle, field strength, energy, or physical configuration under a specified objective and constraint set.
- **Exhaustive enumeration** - A problem asking for the complete set, list, count, or classification of all valid states, modes, solutions, configurations, particles, transitions, paths, quantum numbers, or parameter values satisfying the given criteria.
- **Canonicalization or normalization** - A problem asking for a physical object to be rewritten in a specified standard, reduced, normalized, or canonical form, such as unit-vector notation, SI units, normalized wavefunction, phasor form, standard circuit equivalent, or reduced physical expression.
- **Transformation between representations** - A problem asking for conversion of an object from one physically equivalent representation to another, such as graph to equation, circuit diagram to equations, time domain to frequency domain, Cartesian to polar components, free-body description to force equations, or state vector to measurement probabilities.
- **Parameter identification** - A problem asking for unknown parameters, constants, coefficients, initial conditions, model terms, material properties, circuit values,

- frequencies, rates, or latent physical quantities to be inferred from equations, data, constraints, or observed physical behavior.

Answer Types

[Biology](#)
[Chemistry](#)
[Physics](#)

Biology: Answer Type Taxonomy

Answer type describes the expected format of the final answer, independent of the problem's domain, subdomain, input representation, or problem form. Since answers must be text-only and directly verifiable, answer types should be limited to objects that can be written explicitly using numbers, symbols, labels, finite lists, formulas, names, or compact biological notations. Avoid answer types that require proofs, derivations, diagrams, essays, or subjective explanation.

- **Exact scalar** - A single exact numerical quantity, such as an integer, rational number, count, ratio, probability, percentage, genotype frequency, allele frequency, population size, sequence length, cell count, rate value, or coefficient.
- **Numerical scalar approximation** - A single numerical quantity expected up to a specified tolerance, precision, number of significant figures, number of decimal places, or error bound. Any required tolerance must be stated in the problem and should be tailored towards the problem (e.g., not a blanket 10% allowable tolerance).
- **Exact symbolic expression** - A closed-form symbolic expression representing the correct final answer up to biological or mathematical equivalence, such as a growth equation, inheritance probability formula, population model, rate expression, binding relationship, or regulatory expression.
- **Biological sequence** - A DNA sequence, RNA sequence, amino acid sequence, codon sequence, motif, consensus sequence, primer sequence, alignment segment, or other text-only sequence representation.
- **Genotype or genetic notation** - A genotype, haplotype, allele combination, karyotype notation, mutation notation, linkage notation, pedigree-derived genotype, or inheritance-pattern notation.
- **Biological species or structure** - A gene, protein, cell type, tissue, organ, species, taxon, anatomical structure, pathway component, biomolecule, organelle, or other named biological entity.

- **Quantity with units** - A concentration, rate, mass, volume, length, time, area, density, population size, enzyme activity, expression level, dose, or unit-bearing biological quantity.
- **Tuple or ordered list** - A finite ordered collection of values, sequences, taxa, stages, pathway steps, experimental conditions, labels, or biological states where order matters.

  Set or multiset of objects - A complete finite collection of valid genes, proteins, species, taxa, genotypes, phenotypes, mutations, codons, cell types, structures, stages, or outcomes where order is irrelevant and multiplicity may or may not matter.
- **Sequence or pathway representation** - A developmental sequence, metabolic pathway, signaling cascade, infection cycle, cell-cycle sequence, experimental workflow, lineage path, or ordered biological process.
- **Function, model, or mapping** - A function, growth model, dose-response curve, genotype-phenotype mapping, gene-regulatory mapping, ecological model, binding curve, or transformation rule specified explicitly by formula, table, or action on biological states.
- **Canonical form** - A biological object expressed in a specified standard or normalized form, such as 5′ to 3′ sequence orientation, standard amino acid code, normalized expression value, binomial species name, HGVS-style mutation notation, or standardized genotype notation.
- **Classification label** - A category, class, type, case, biological state, inheritance mode, taxonomic group, cell type, tissue type, reaction class, ecological interaction, developmental stage, disease mechanism, or qualitative behavior label.

Chemistry: Answer Type Taxonomy

Answer type describes the expected format of the final answer, independent of the problem's domain, subdomain, input representation, or problem form. Since answers must be text-only and directly verifiable, answer types should be limited to objects that can be written explicitly using numbers, symbols, labels, finite lists, formulas, names, or compact structural notations. Avoid answer types that require proofs, derivations, diagrams, essays, or subjective explanation.

- **Exact scalar** - A single exact numerical quantity, such as an integer, rational number, stoichiometric coefficient, mole quantity, charge, oxidation state, particle count, equilibrium constant, rate constant, enthalpy value, free-energy value, or coefficient.
- **Numerical scalar approximation** - A single numerical quantity expected up to a specified tolerance, precision, number of significant figures, number of decimal places, or error bound. Any required tolerance must be stated in the problem and should be tailored towards the problem (e.g., not a blanket 10% allowable tolerance).
- **Exact symbolic expression** - A closed-form symbolic expression representing the correct final answer up to chemical or mathematical equivalence, such as a rate law, equilibrium expression, thermodynamic relationship, concentration expression, integrated law, or formula.

- **Chemical formula or composition** - A molecular formula, empirical formula, ionic formula, hydrate formula, isotope notation, elemental composition, percent composition, or formula-unit representation.
- **Balanced chemical equation** - A molecular equation, ionic equation, net ionic equation, redox equation, half-reaction, combustion equation, or reaction scheme written explicitly in text.
- **Chemical species or structure** - A chemical name, ion, molecule, product, intermediate, reagent, catalyst, ligand, complex, functional group, condensed structural formula, SMILES string, InChI string, or other text-only structural representation.

- **Concentration or quantity with units** - A mole amount, mass, volume, pressure, temperature, concentration, molarity, molality, mole fraction, ppm value, charge, current, time, or unit-bearing chemical quantity.
- **Tuple or ordered list** - A finite ordered collection of values, species, coefficients, products, reagents, conditions, labels, peaks, steps, or coordinates where order matters.
- **Set or multiset of objects** - A complete finite collection of valid species, isomers, products, resonance forms, phases, complexes, peaks, reagents, or configurations where order is irrelevant and multiplicity may or may not matter.
- **Sequence or pathway representation** - A reaction sequence, synthetic route, mechanism step sequence, metabolic pathway segment, separation sequence, recurrence-like process, or ordered transformation chain.
- **Function, law, or mapping** - A function, calibration curve, rate law, response factor, partition relationship, binding curve, transformation rule, or mapping specified explicitly by formula, table, or action on species.
- **Canonical form** - A chemical object expressed in a specified standard or normalized form, such as a balanced equation, reduced empirical formula, IUPAC name, standard cell notation, canonical SMILES, oxidation-state-normalized formula, or standardized equilibrium expression.
- **Classification label** - A category, class, type, case, structural classification, reaction type, mechanism type, functional group class, stereochemical label, phase label, acid-base classification, redox classification, or qualitative behavior label.

Physics: Answer Type Taxonomy

Answer type describes the expected format of the final answer, independent of the problem's domain, subdomain, input representation, or problem form. Since answers must be text-only and directly verifiable, answer types should be limited to objects that can be written explicitly using numbers, symbols, labels, finite lists, formulas, names, or compact physical notations. Avoid answer types that require proofs, derivations, diagrams, essays, or subjective explanation.

- **Exact scalar** - A single exact numerical quantity, such as an integer, rational number, coefficient, count, quantum number, frequency ratio, efficiency, probability, dimensionless constant, or exact physical value.
- **Numerical scalar approximation** - A single numerical quantity expected up to a specified tolerance, precision, number of significant figures, number of decimal places, or error bound. Any required tolerance must be stated in the problem and should be tailored towards the problem (e.g., not a blanket 10% allowable tolerance).
- **Exact symbolic expression** - A closed-form symbolic expression representing the correct final answer up to physical or mathematical equivalence, such as an equation of motion, field expression, energy expression, wavefunction, transfer function, scaling law, or physical relationship.

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- **Physical quantity with units** - A force, energy, power, pressure, temperature, current, voltage, resistance, capacitance, field strength, wavelength, frequency, speed, mass, time, distance, or other unit-bearing physical quantity.
- **Equation or physical law** - An equation, conservation statement, field equation, boundary condition, dispersion relation, circuit equation, wave equation, or constraint written explicitly as the answer.

  Vector - A finite-dimensional vector returned as components, unit-vector notation, coordinates, momentum, velocity, force, field vector, or state vector.
- **Matrix, tensor, or operator** - A matrix, tensor, operator, inertia tensor, stress tensor, Hamiltonian, transformation matrix, or other finite-dimensional linear object written explicitly.
- **Interval, range, or region** - A time interval, spatial region, frequency range, wavelength band, allowed parameter range, domain of validity, or inequality-defined physical region.
- **Tuple or ordered list** - A finite ordered collection of values, states, labels, modes, particles, transitions, steps, coordinates, or parameters where order matters.
- **Set or multiset of objects** - A complete finite collection of valid states, modes, solutions, configurations, particles, transitions, quantum numbers, paths, or classifications where order is irrelevant and multiplicity may or may not matter.
- **Function, field, or model** - A function, trajectory, wavefunction, potential, field, distribution, response curve, transfer function, Green's function, or model specified explicitly by formula, table, or action on variables.
- **Sequence or process representation** - A sequence of states, transitions, decay steps, collision outcomes, circuit states, measurement outcomes, or time-ordered physical events.
- **Canonical form** - A physical object expressed in a specified standard or normalized form, such as SI-unit form, unit-vector form, normalized wavefunction, phasor form, standard circuit equivalent, diagonalized operator, or reduced expression.
- **Classification label** - A category, class, type, case, physical regime, phase, particle type, interaction type, stability label, motion type, wave mode, circuit behavior, or qualitative behavior label.

Domain Reference

Domain and Sub-Domain Descriptions

[Biology](#)
[Chemistry](#)
[Physics](#)

Biology

Cell Biology

Problems involving cellular structures, organelles, membranes, cell division, intracellular transport, and cellular regulation. Avoiding and excluding broad organism-level physiology, ecology, or molecular-genetic mechanisms unless the primary task is cellular structure or function.

- **Cell structure and organelles** - Identifying or comparing nuclei, mitochondria, chloroplasts, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, vacuoles, cytoskeleton, and other cellular components.
- **Membrane structure and transport** - Working with phospholipid bilayers, channels, pumps, diffusion, facilitated diffusion, osmosis, active transport, endocytosis, and exocytosis.
- **Cell cycle and mitosis** - Identifying cell-cycle phases, checkpoints, mitotic stages, chromosome behavior, and daughter-cell outcomes.
- **Meiosis and gametogenesis** - Working with meiotic stages, crossing over, independent assortment, haploid products, nondisjunction, and gamete formation.
- **Cytoskeleton and cell motility** - Analyzing microtubules, actin filaments, intermediate filaments, motor proteins, cilia, flagella, and intracellular movement.
- **Cell communication** - Working with receptors, ligands, second messengers, signal transduction, feedback, and cellular responses.
- **Cell differentiation and specialization** - Identifying differentiated cell types, stem cells, potency, lineage restriction, and specialized cellular functions.
- **Cell death and survival** - Working with apoptosis, necrosis, autophagy, senescence, and survival signaling.

Molecular Biology

Problems involving DNA, RNA, gene expression, replication, transcription, translation, and molecular mechanisms of biological information flow. Avoiding and excluding population-level genetics, evolutionary inference, or broad genomics unless the central task is molecular mechanism or sequence-level interpretation.

- **DNA structure and replication** - Working with nucleotide structure, base pairing, antiparallel strands, replication forks, polymerases, primers, Okazaki fragments, and replication fidelity.

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  Transcription - Identifying promoters, templates, RNA polymerase activity, transcription direction, RNA products, and regulatory elements.

  RNA processing - Working with splicing, introns, exons, capping, polyadenylation, editing, and mature transcript formation.

  Translation - Using codons, anticodons, reading frames, ribosomes, tRNAs, start and stop codons, and amino acid sequences.

  Gene regulation - Working with operons, enhancers, silencers, transcription factors, epigenetic marks, chromatin accessibility, and regulatory networks.

  Mutations and molecular consequences - Classifying substitutions, insertions, deletions, frameshifts, missense, nonsense, synonymous, splice-site, and regulatory mutations.
- **Molecular techniques** - Interpreting PCR, gel electrophoresis, cloning, restriction digestion, sequencing, qPCR, and blotting outputs.
- **Genome editing** - Working with CRISPR-Cas systems, guide RNAs, target sites, edits, off-target considerations, and repair pathways.

Genetics and Genomics

Problems involving inheritance, genetic variation, linkage, pedigrees, chromosomes, genomes, and genotype-phenotype relationships. Avoiding and excluding molecular mechanism details or evolutionary population dynamics unless the primary task is genetic inheritance or genomic interpretation.

- **Mendelian inheritance** - Solving monohybrid, dihybrid, dominance, recessiveness, segregation, independent assortment, and genotype-phenotype probability problems.
- **Non-Mendelian inheritance** - Working with incomplete dominance, codominance, epistasis, penetrance, expressivity, pleiotropy, genomic imprinting, and mitochondrial inheritance.
- **Pedigree analysis** - Inferring inheritance patterns, carrier probabilities, genotypes, affected status, and recurrence risks from family data.
- **Linkage and recombination** - Computing recombination frequency, genetic distance, linkage phase, map order, and crossover outcomes.
- **Chromosomal abnormalities** - Identifying aneuploidy, deletions, duplications, inversions, translocations, nondisjunction, and karyotype results.
- **Quantitative genetics** - Working with polygenic traits, heritability, variance components, breeding values, and genotype-environment effects.
- **Genomics** - Interpreting genome organization, gene annotation, copy number variation, genome size, synteny, and comparative genomic features.
- **Population genetics** - Computing allele frequencies, genotype frequencies, Hardy-Weinberg expectations, selection coefficients, drift outcomes, and migration effects.

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Evolutionary Biology

Problems involving descent with modification, natural selection, phylogeny, adaptation, speciation, and evolutionary change. Avoiding and excluding open-ended historical narratives or subjective adaptation explanations unless the expected answer is a classification, value, tree, ordering, or directly checkable evolutionary inference.

- **Natural selection** - Working with fitness, selection pressures, selection coefficients, adaptive traits, and changes in allele or phenotype frequency.

  Genetic drift and gene flow - Computing or identifying founder effects, bottlenecks, migration effects, random allele-frequency change, and population structure.

  Mutation and variation - Working with mutation rates, sources of variation, neutral variation, deleterious mutations, and beneficial variants.

  Phylogenetics - Interpreting trees, clades, sister taxa, common ancestors, character states, branch lengths, and phylogenetic relationships.

  Speciation - Classifying allopatric, sympatric, parapatric, and peripatric speciation, reproductive isolation, and hybrid zones.
- **Evolutionary mechanisms** - Comparing selection, drift, mutation, recombination, migration, and nonrandom mating as drivers of change.
- **Molecular evolution** - Working with substitution rates, molecular clocks, conserved sequences, homologs, orthologs, paralogs, and sequence divergence.
- **Adaptation and life-history evolution** - Analyzing tradeoffs, reproductive strategies, survival, fecundity, sexual selection, and adaptive constraints.

Ecology

Problems involving interactions among organisms and environments, populations, communities, ecosystems, and ecological processes. Avoiding and excluding molecular, cellular, or organismal physiology unless the primary task is ecological structure, interaction, or quantitative ecological reasoning.

- **Population ecology** - Computing growth rates, carrying capacity, survivorship, age structure, density dependence, and population projections.
- **Community ecology** - Working with species richness, diversity indices, competition, predation, mutualism, parasitism, succession, and community composition.
- **Ecosystem ecology** - Analyzing energy flow, trophic levels, food webs, productivity, decomposition, and nutrient cycling.
- **Behavioral ecology** - Interpreting foraging, mating systems, parental investment, cooperation, competition, and cost-benefit tradeoffs.
- **Conservation biology** - Working with extinction risk, habitat fragmentation, biodiversity metrics, population viability, invasive species, and conservation prioritization.
- **Biogeography** - Interpreting species distributions, dispersal, endemism, island biogeography, range shifts, and environmental gradients.

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- **Ecological modeling** - Applying exponential growth, logistic growth, predator-prey models, metapopulation models, and ecological simulations.
- **Human impacts and global change** - Analyzing climate effects, land use, pollution, eutrophication, overharvesting, and ecological disturbance.

Physiology and Homeostasis

Problems involving organismal function, homeostatic regulation, organ systems, and physiological responses. Avoiding and excluding detailed anatomy-only identification or molecular signaling unless the primary task is system function, regulation, or physiological calculation.

Homeostasis and feedback - Working with negative feedback, positive feedback, set points, regulation loops, and physiological control systems.

Cardiovascular physiology - Analyzing heart function, blood flow, blood pressure, cardiac output, gas transport, and circulatory regulation.

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  Respiratory physiology - Working with ventilation, gas exchange, oxygen transport, carbon dioxide transport, diffusion gradients, and respiratory control.

  Renal and osmoregulatory physiology - Computing or identifying filtration, reabsorption, secretion, osmolarity, urine concentration, and water-salt balance.

  Digestive physiology and nutrition - Working with digestion, absorption, enzymes, nutrient processing, energy balance, and gastrointestinal regulation.

  Endocrine physiology - Identifying hormones, glands, target tissues, feedback loops, and endocrine regulation.
- **Musculoskeletal physiology** - Working with muscle contraction, sarcomeres, motor units, skeletal function, and movement.
- **Thermoregulation and stress physiology** - Analyzing temperature regulation, metabolic rate, acclimation, stress hormones, and physiological responses to environmental change.

Anatomy and Morphology

Problems involving biological structures, body plans, tissues, organs, morphology, and structural comparison. Avoiding and excluding physiological mechanisms unless the primary task is anatomical identification, structural relationship, or morphological classification.

- **Human anatomy** - Identifying organs, tissues, anatomical regions, body systems, planes, cavities, and structural relationships.
- **Comparative anatomy** - Comparing homologous, analogous, vestigial, and derived structures across taxa.
- **Tissue structure** - Working with epithelial, connective, muscle, nervous, vascular, xylem, phloem, and other tissue types.
- **Organ systems** - Identifying components and structural organization of circulatory, respiratory, digestive, nervous, endocrine, reproductive, immune, and excretory systems.
- **Morphological traits** - Classifying body forms, appendages, segmentation, symmetry, skeletal structures, and external features.
- **Plant anatomy** - Working with roots, stems, leaves, flowers, meristems, vascular tissues, stomata, and reproductive structures.
- **Developmental morphology** - Identifying structural changes across developmental stages, metamorphosis, embryonic structures, and patterning outcomes.
- **Anatomical imaging and diagrams** - Interpreting labeled diagrams, imaging outputs, cross-sections, and structural schematics.

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Developmental Biology

Problems involving embryogenesis, cell fate, pattern formation, morphogenesis, and developmental regulation. Avoiding and excluding broad organismal life-cycle description unless the primary task is developmental mechanism, stage identification, or directly checkable developmental outcome.

- **Gametogenesis and fertilization** - Working with gamete formation, fertilization, zygote formation, polarity, and early developmental activation.

  Cleavage and gastrulation - Identifying early embryonic stages, blastula formation, germ layers, gastrulation movements, and fate maps.

  Organogenesis - Working with tissue differentiation, organ formation, developmental timing, and structural development.

  Cell fate and differentiation - Interpreting lineage restriction, potency, induction, determination, and differentiation markers.

  Pattern formation - Working with morphogens, gradients, segmentation, body axes, Hox genes, and positional information.

  Developmental gene regulation - Identifying regulatory genes, signaling pathways, transcription factors, and gene-expression patterns during development.

  Stem cells and regeneration - Working with embryonic stem cells, adult stem cells, induced pluripotency, regeneration, and lineage tracing.
- **Developmental disorders** - Identifying effects of mutations, teratogens, chromosomal abnormalities, and disrupted developmental pathways.

Microbiology and Virology

Problems involving bacteria, archaea, viruses, microbial growth, microbial genetics, infection cycles, and microbial diversity. Avoiding and excluding human immune response or clinical diagnosis unless the primary task is microbial identity, growth, replication, or mechanism.

- **Bacterial cell structure** - Identifying cell walls, membranes, capsules, flagella, pili, plasmids, spores, and Gram-positive or Gram-negative features.
- **Microbial growth** - Computing growth rates, doubling time, growth phases, culture density, and environmental effects on microbial populations.
- **Microbial metabolism** - Working with aerobic respiration, anaerobic respiration, fermentation, chemotrophy, phototrophy, and metabolic classification.
- **Microbial genetics** - Interpreting transformation, transduction, conjugation, plasmids, horizontal gene transfer, and antibiotic resistance genes.
- **Virology** - Working with viral structure, genome type, host range, replication cycles, lytic and lysogenic cycles, and viral classification.
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- **Pathogenesis and virulence** - Identifying toxins, adhesion, invasion, virulence factors, infection routes, and host-pathogen interactions.
- **Antimicrobials and resistance** - Working with antibiotic mechanisms, resistance patterns, susceptibility data, MIC values, and treatment constraints.
- **Microbial ecology** - Analyzing microbiomes, biofilms, symbiosis, nutrient cycling, and microbial community interactions.

Immunology

Problems involving immune cells, immune responses, antibodies, antigens, inflammation, vaccination, and immune dysfunction. Avoiding and excluding general microbiology or physiology unless the central task is immune recognition, response, regulation, or classification.

- **Innate immunity** - Identifying barriers, phagocytes, complement, inflammation, pattern-recognition receptors, cytokines, and innate immune responses.
- **Adaptive immunity** - Working with B cells, T cells, antigen presentation, clonal selection, memory cells, and adaptive response timing.
- **Antibodies and antigens** - Interpreting antibody classes, antigen binding, epitopes, neutralization, opsonization, and antibody-mediated effects.

  Major histocompatibility complex - Working with MHC class I, MHC class II, antigen presentation, T-cell recognition, and tissue compatibility.

  Vaccination and immunological memory - Analyzing vaccine types, boosters, primary and secondary responses, herd immunity, and memory-cell behavior.

  Hypersensitivity and autoimmunity - Classifying allergy, hypersensitivity types, autoimmune mechanisms, and tolerance breakdown.

  Immunodeficiency - Identifying primary and acquired immune defects, affected cell types, and expected infection patterns.

  Immunological assays - Interpreting ELISA, flow cytometry, Western blot, agglutination, neutralization assays, and serological data.

Neuroscience and Behavior

Problems involving neurons, nervous systems, sensory processing, behavior, and neural signaling. Avoiding and excluding broad psychology or open-ended behavioral interpretation unless the expected answer is a mechanism, classification, value, pathway, or directly checkable prediction.

- **Neuron structure and function** - Identifying dendrites, axons, soma, myelin, synapses, glia, and neuron types.
- **Membrane potentials and action potentials** - Working with resting potential, depolarization, repolarization, refractory periods, ion gradients, and conduction.

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- **Synaptic transmission** - Interpreting neurotransmitters, receptors, synaptic vesicles, postsynaptic potentials, and synaptic integration.
- **Sensory systems** - Working with photoreception, mechanoreception, chemoreception, auditory processing, visual pathways, and sensory transduction.
- **Motor systems** - Analyzing motor neurons, neuromuscular junctions, reflex arcs, motor control, and muscle activation.
- **Brain regions and circuits** - Identifying major brain structures, functional regions, neural pathways, and circuit organization.
- **Learning and plasticity** - Working with synaptic plasticity, long-term potentiation, habituation, sensitization, conditioning, and memory-related changes.
- **Behavioral biology** - Interpreting innate behavior, learned behavior, communication, social behavior, circadian rhythms, and behavioral experiments.

Plant Biology

Problems involving plant structure, physiology, reproduction, growth, signaling, and adaptation. Avoiding and excluding general ecology or agricultural discussion unless the primary task is plant-specific biological structure, function, or response.

- **Plant cells and tissues** - Identifying cell walls, chloroplasts, vacuoles, xylem, phloem, meristems, epidermis, cortex, and vascular tissues.
- **Photosynthesis** - Working with light reactions, Calvin cycle, pigments, carbon fixation, stomata, and photosynthetic rates.
- **Plant water and nutrient transport** - Analyzing transpiration, cohesion-tension, root uptake, mineral nutrition, phloem transport, and source-sink relationships.
- **Plant growth and hormones** - Working with auxin, gibberellin, cytokinin, abscisic acid, ethylene, tropisms, and growth regulation.
- **Plant reproduction** - Identifying flowers, pollen, ovules, seeds, fruits, alternation of generations, fertilization, and reproductive cycles.
- **Plant responses to environment** - Interpreting photoperiodism, gravitropism, phototropism, drought response, defense response, and stress adaptation.

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  Plant diversity and classification - Classifying bryophytes, seedless vascular plants, gymnosperms, angiosperms, monocots, and eudicots.

  Plant biotechnology - Working with transformation, tissue culture, genetically modified plants, marker selection, and crop trait analysis.

Biochemistry and Metabolism

Problems involving biomolecules, enzymes, metabolic pathways, bioenergetics, and chemical processes in living systems. Avoiding and excluding purely chemical thermodynamics or organic synthesis unless the primary task is biological function, pathway logic, or biochemical calculation.

- **Biomolecules** - Identifying structures and properties of carbohydrates, lipids, proteins, nucleic acids, cofactors, and metabolites.
- **Protein structure and function** - Working with amino acid properties, peptide bonds, primary through quaternary structure, folding, binding, and denaturation.
- **Enzyme kinetics** - Computing Michaelis-Menten quantities, Km, Vmax, kcat, catalytic efficiency, inhibition type, and reaction rates.
- **Cellular respiration** - Working with glycolysis, pyruvate oxidation, citric acid cycle, electron transport, oxidative phosphorylation, and ATP yields.
- **Fermentation and anaerobic metabolism** - Identifying fermentation products, NAD+ regeneration, oxygen-independent pathways, and energy yields.
- **Photosynthetic metabolism** - Working with carbon fixation, RuBisCO, C3, C4, CAM pathways, and photorespiration.
- **Metabolic regulation** - Interpreting feedback inhibition, allosteric regulation, covalent modification, pathway control, and energy-state effects.
- **Bioenergetics** - Computing or comparing ATP coupling, redox carriers, free energy changes, proton gradients, and thermodynamic favorability.

Bioinformatics and Computational Biology

Problems involving biological sequence analysis, omics data, computational models, databases, and algorithmic biological interpretation. Avoiding and excluding open-ended software discussion unless the expected answer is a directly checkable output, classification, parameter, sequence, alignment, or numerical result.

- **Sequence alignment** - Working with pairwise alignments, multiple alignments, scoring matrices, gaps, identity, similarity, and alignment scores.
- **Genome annotation** - Identifying open reading frames, genes, exons, introns, promoters, regulatory elements, and coding regions.
- **Phylogenomic analysis** - Interpreting sequence-based trees, orthologs, paralogs, conserved regions, and comparative genomic relationships.
- **Transcriptomics** - Working with RNA-seq counts, expression levels, fold change, differential expression, normalization, and clustering.
- **Proteomics** - Interpreting peptide matches, protein abundance, post-translational modifications, mass spectrometry outputs, and protein identification.

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- **Biological databases and identifiers** - Mapping genes, proteins, accessions, annotations, ontology terms, and database-derived records.

  Network biology - Working with gene regulatory networks, protein interaction networks, pathway graphs, centrality, and modules.

  Computational modeling - Applying Boolean models, differential equation models, stochastic models, simulations, and parameterized biological systems.

Experimental Biology and Biostatistics

Problems involving experimental design, biological data interpretation, measurement, controls, statistical inference, and reproducible analysis. Avoiding and excluding open-ended critique or essay-style methodology unless the expected answer is a classification, calculation, selected design, statistical value, or directly checkable interpretation.

- **Experimental design** - Identifying independent variables, dependent variables, controls, replication, randomization, blinding, and treatment groups.
- **Microscopy and imaging** - Interpreting magnification, resolution, staining, fluorescence, image scale bars, and cellular or tissue images.
- **Laboratory assays** - Working with ELISA, Western blot, PCR, qPCR, gels, reporter assays, enzyme assays, growth assays, and cell-counting assays.
- **Data visualization** - Interpreting tables, graphs, scatter plots, dose-response curves, survival curves, electropherograms, and heatmaps.
- **Biostatistical inference** - Computing p-values, confidence intervals, test statistics, effect sizes, sample sizes, and significance decisions.
- **Experimental controls and validity** - Identifying positive controls, negative controls, confounders, artifacts, bias, contamination, and validity threats.
- **Dose-response and response curves** - Computing EC50, IC50, LD50, Hill coefficients, slopes, thresholds, and response classifications.
- **Measurement error and uncertainty** - Working with standard deviation, standard error, coefficient of variation, error bars, technical replicates, and biological replicates.

Taxonomy and Biodiversity

Problems involving classification, organismal diversity, phylogenetic groups, life histories, and biological identification. Avoiding and excluding ecology or evolution as the primary task unless the answer concerns taxonomic placement, diagnostic traits, or biodiversity classification.

- **Biological classification** - Working with domains, kingdoms, phyla, classes, orders, families, genera, species, and taxonomic hierarchy.
- **Species concepts** - Applying biological, morphological, phylogenetic, ecological, and genetic species concepts.
- **Prokaryotic diversity** - Classifying bacteria and archaea by structure, metabolism, habitat, and phylogenetic group.
- **Protist and fungal diversity** - Identifying major groups, life cycles, nutrition modes, reproductive features, and ecological roles.
- **Animal diversity** - Classifying major animal phyla, body plans, symmetry, development, segmentation, and diagnostic traits.

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- **Plant diversity** - Classifying plant groups, vascular tissue, seeds, flowers, spores, and reproductive structures.
- **Biodiversity metrics** - Computing species richness, abundance, evenness, diversity indices, and similarity metrics.
- **Identification keys and diagnostic traits** - Using dichotomous keys, trait matrices, morphological characters, and genetic markers to identify organisms.

Chemistry

General Chemistry and Measurement

Problems involving foundational chemical quantities, units, measurement, dimensional analysis, and introductory chemical reasoning. Avoiding and excluding advanced mechanistic, quantum, thermodynamic, or analytical interpretation unless the primary task is basic chemical calculation or representation.

- **Units, measurement, and significant figures** - Working with SI units, unit conversions, dimensional analysis, uncertainty, precision, accuracy, and significant figures.
- **Moles and molar quantities** - Computing moles, molar mass, particle counts, Avogadro's number, empirical quantities, and formula-unit relationships.
- **Chemical formulas and nomenclature** - Interpreting, naming, and writing formulas for ionic compounds, molecular compounds, acids, bases, hydrates, and simple coordination compounds.
- **Chemical composition** - Computing mass percent, percent composition, empirical formulas, molecular formulas, and elemental composition.
- **Laboratory quantities and preparation** - Computing masses, volumes, dilutions, concentrations, yields, and reagent quantities for standard laboratory preparation.
- **Chemical classification** - Classifying substances as elements, compounds, mixtures, electrolytes, acids, bases, salts, oxidants, reductants, or reaction types.
- **Introductory chemical modeling** - Representing simple chemical systems using formulas, equations, particulate descriptions, or proportional relationships.

Atomic Structure and Periodicity

Problems involving electronic structure, atomic models, periodic trends, isotopes, and atomic-scale properties. Avoiding and excluding detailed molecular orbital theory or quantum-mechanical derivations unless the final answer is a directly checkable configuration, value, label, or ordering.

- **Atomic structure** - Working with protons, neutrons, electrons, isotopes, ions, mass number, atomic number, and average atomic mass.
- **Electron configurations** - Writing electron configurations, orbital box diagrams, valence configurations, and noble-gas shorthand.
- **Quantum numbers and orbitals** - Identifying allowed quantum numbers, orbital labels, subshell capacities, and electron occupancy.

- **Periodic trends** - Comparing atomic radius, ionic radius, ionization energy, electron affinity, electronegativity, metallic character, and effective nuclear charge.
- **Isotopes and mass spectrometry** - Computing average atomic mass, isotope abundance, mass-to-charge interpretation, and isotopic patterns.
- **Photoelectron and atomic spectra** - Interpreting ionization-energy data, atomic transitions, emission wavelengths, and energy-level relationships.
- **Periodic classification** - Classifying elements by group, period, block, valence shell, chemical family, and expected reactivity pattern.

Chemical Bonding and Molecular Structure

Problems involving bonds, molecular geometry, polarity, intermolecular forces, and structure-property relationships. Avoiding and excluding long structural explanations or molecular-orbital derivations unless the expected answer is a structure, label, ordering, or computed property.

- **Lewis structures** - Drawing or identifying valid Lewis structures, formal charges, resonance forms, octets, radicals, and expanded valence cases.
- **VSEPR and molecular geometry** - Determining electron-domain geometry, molecular shape, bond angles, lone-pair effects, and hybridization.
- **Bond polarity and molecular polarity** - Determining dipoles, molecular polarity, partial charges, and polarity-based orderings.
- **Intermolecular forces** - Classifying and comparing dispersion forces, dipole-dipole interactions, hydrogen bonding, ion-dipole interactions, and related effects.
- **Bond order and bond strength** - Computing or comparing bond order, bond length, bond energy, and resonance-based bond equivalence.
- **Valence bond and hybridization models** - Identifying sigma and pi bonds, hybrid orbitals, orbital overlap, and local bonding descriptions.
- **Molecular orbital theory** - Working with bonding and antibonding orbitals, MO diagrams, magnetism, bond order, and electron occupancy.
- **Structure-property relationships** - Relating molecular structure to boiling point, melting point, solubility, viscosity, volatility, acidity, or reactivity.

Stoichiometry and Chemical Reactions

Problems involving balanced chemical equations, reaction quantities, limiting reagents, yields, and reaction classification. Avoiding and excluding kinetic, equilibrium, or thermodynamic interpretation unless the primary task is stoichiometric calculation or reaction accounting.

- **Balancing chemical equations** - Balancing molecular, ionic, redox, combustion, precipitation, and acid-base equations.
- **Reaction stoichiometry** - Computing masses, moles, volumes, particle counts, and reactant-product quantities from balanced equations.
- **Limiting reactants and excess reagents** - Identifying limiting reagents, excess amounts, theoretical yields, and remaining reactant quantities.
- **Percent yield and purity** - Computing actual yield, theoretical yield, percent yield, percent purity, and recovery.
- **Reaction classification** - Classifying synthesis, decomposition, combustion, single-replacement, double-replacement, precipitation, acid-base, and redox reactions.
- **Net ionic equations** - Writing complete ionic and net ionic equations, spectator ions, and precipitate formation.
- **Combustion analysis** - Determining empirical or molecular formulas from combustion products and mass data.
- **Gas stoichiometry** - Combining balanced equations with gas volumes, molar volume, partial pressures, or ideal gas relationships.

States of Matter and Phase Behavior

Problems involving gases, liquids, solids, phase changes, and macroscopic physical behavior of substances. Avoiding and excluding solution chemistry, thermodynamic derivations, or materials-science structure unless the primary task is phase or state-property calculation.

- **Gas laws** - Applying Boyle's law, Charles's law, Avogadro's law, combined gas law, ideal gas law, and gas density relationships.
- **Gas mixtures and partial pressures** - Computing mole fractions, partial pressures, total pressures, collected-gas corrections, and gas composition.
- **Real gases** - Applying van der Waals corrections, compressibility, deviations from ideality, and qualitative real-gas behavior.
- **Liquids and solids** - Comparing viscosity, surface tension, vapor pressure, melting point, boiling point, and crystal versus amorphous structure.
- **Phase changes** - Computing heat of fusion, heat of vaporization, phase-transition quantities, and heating or cooling curve values.
- **Phase diagrams** - Interpreting phase boundaries, triple points, critical points, normal boiling points, and stable phases.
- **Crystalline solids** - Working with unit cells, packing, coordination number, density, lattice parameters, and crystal structures.
- **Colligative phase behavior** - Computing boiling-point elevation, freezing-point depression, osmotic pressure, and vapor-pressure lowering.

Solutions and Mixtures

Problems involving concentration, solubility, dilution, solution preparation, and mixture behavior. Avoiding and excluding advanced aqueous equilibria or analytical-instrument interpretation unless the primary task is solution calculation.

- **Concentration units** - Computing molarity, molality, mole fraction, mass percent, volume percent, ppm, ppb, and normality.
- **Dilution and mixing** - Solving dilution, serial dilution, mixture, and concentration-change problems.
- **Solubility and saturation** - Determining saturated, unsaturated, and supersaturated states, solubility quantities, and temperature-dependent solubility.
- **Electrolytes and solution conductivity** - Classifying strong, weak, and non-electrolytes and computing ion concentrations from dissociation.
- **Colligative properties** - Computing osmotic pressure, freezing-point depression, boiling-point elevation, and molar mass from solution behavior.
- **Activity and non-ideal solutions** - Working with activity coefficients, ionic strength, Debye-Hückel approximations, and deviations from ideality.
- **Partitioning and extraction** - Computing distribution ratios, extraction efficiency, and solute partitioning between phases.
- **Solvent and solute effects** - Comparing polarity, miscibility, solvation, hydration, and qualitative solubility patterns.

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Chemical Thermodynamics

Problems involving heat, energy, entropy, enthalpy, Gibbs free energy, spontaneity, and thermodynamic state functions. Avoiding and excluding kinetic explanations or statistical-mechanical derivations unless the final answer is a directly checkable thermodynamic value or classification.

Thermochemical calculations - Computing heat, specific heat, calorimetry values, heat capacity, and temperature changes.

Enthalpy and Hess's law - Computing reaction enthalpies from formation enthalpies, bond energies, calorimetry, and Hess cycles.

Entropy - Computing or comparing entropy changes, standard molar entropies, and qualitative entropy trends.

Gibbs free energy - Computing free energy, spontaneity, equilibrium relationships, and temperature-dependent feasibility.

- **Thermodynamic state functions** - Working with internal energy, enthalpy, entropy, Gibbs energy, path independence, and state changes.
- **Phase-transition thermodynamics** - Computing thermodynamic quantities for fusion, vaporization, sublimation, and phase equilibria.
- **Temperature dependence** - Applying Kirchhoff-type relationships, van 't Hoff relationships, and temperature effects on thermodynamic quantities.
- **Statistical thermodynamics** - Computing partition-function-based quantities, populations, Boltzmann factors, and molecular contributions to thermodynamic properties.

Chemical Kinetics

Problems involving reaction rates, rate laws, mechanisms, activation energy, and time-dependent concentration behavior. Avoiding and excluding equilibrium-only or thermodynamic-only interpretation unless the primary task is rate-based calculation or mechanism inference.

- **Reaction rates** - Computing average rates, instantaneous rates, stoichiometric rate relationships, and concentration-time changes.
- **Rate laws and reaction order** - Determining rate laws, orders, rate constants, and units from data or mechanisms.
- **Integrated rate laws** - Applying zero-order, first-order, second-order, and pseudo-first-order concentration-time relationships.
- **Half-life and lifetime** - Computing half-lives, decay times, reaction completion times, and concentration thresholds.
- **Arrhenius behavior** - Computing activation energy, frequency factors, temperature effects, and rate-constant ratios.
- **Reaction mechanisms** - Identifying elementary steps, intermediates, catalysts, molecularity, and rate-determining steps.

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- **Steady-state and pre-equilibrium approximations** - Applying approximation-based mechanism analysis to obtain rate laws or concentrations.
- **Catalysis kinetics** - Working with homogeneous catalysis, heterogeneous catalysis, enzyme kinetics, Michaelis-Menten quantities, and catalytic efficiency.

Chemical Equilibrium

Problems involving reversible reactions, equilibrium constants, reaction quotients, equilibrium composition, and perturbations. Avoiding and excluding acid-base-specific or electrochemical equilibria unless those are secondary to general equilibrium reasoning.

Equilibrium constants - Computing Kc, Kp, Kx, equilibrium expressions, and relationships among constants.

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  Reaction quotient and direction - Computing Q and determining reaction direction relative to equilibrium.

  Equilibrium composition - Solving ICE-table problems, equilibrium concentrations, partial pressures, and extent of reaction.

  Le Châtelier's principle - Predicting shifts from concentration, pressure, volume, temperature, and removal or addition changes.

  Heterogeneous equilibria - Working with solids, liquids, gases, and equilibrium expressions for heterogeneous systems.
- **Coupled equilibria** - Combining equilibrium constants, reversing reactions, scaling reactions, and adding reactions.
- **Temperature dependence of equilibrium** - Applying thermodynamic relationships between equilibrium constants, enthalpy, entropy, and temperature.
- **Equilibrium approximations** - Applying small-x approximations, checking approximation validity, and solving resulting equilibrium equations.

Acids, Bases, and Aqueous Equilibria

Problems involving acid-base reactions, pH, buffers, titrations, solubility equilibria, and aqueous ionic systems. Avoiding and excluding general equilibrium problems where acid-base or solubility chemistry is not the main concept.

- **Acid-base definitions and strength** - Classifying Arrénius, Brønsted-Lowry, and Lewis acids or bases and comparing acid or base strength.
- **pH and pOH calculations** - Computing pH, pOH, hydronium concentration, hydroxide concentration, and related quantities.
- **Weak acid and weak base equilibria** - Solving Ka, Kb, percent ionization, conjugate-pair, and equilibrium-concentration problems.
- **Buffers** - Applying Henderson-Hasselbalch relationships, buffer capacity, buffer preparation, and pH changes after additions.
- **Acid-base titrations** - Computing equivalence points, half-equivalence points, titration pH values, indicators, and titration curves.
- **Polyprotic acids and bases** - Working with stepwise dissociation, dominant species, and pH for polyprotic systems.
- **Solubility equilibria** - Computing Ksp, molar solubility, common-ion effects, selective precipitation, and precipitation thresholds.
- **Complex-ion equilibria** - Working with formation constants, ligand binding, complex stability, and solubility changes from complexation.

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Electrochemistry

Problems involving oxidation-reduction, electrochemical cells, electrode potentials, electrolysis, and electrochemical thermodynamics. Avoiding and excluding redox-only stoichiometry unless the primary task is electrochemical behavior or cell calculation.

- **Oxidation states and redox classification** - Assigning oxidation numbers, identifying oxidants and reductants, and classifying redox processes.
- **Balancing redox reactions** - Balancing oxidation-reduction reactions in acidic, basic, or neutral media.
- **Galvanic cells** - Computing cell potentials, identifying anodes and cathodes, writing cell notation, and predicting spontaneous direction.

  Standard reduction potentials - Using reduction tables to compute E°cell, rank oxidizing or reducing agents, and determine feasibility.

  Nernst equation - Computing nonstandard cell potentials, concentration effects, and equilibrium constants from electrochemical data.

  Electrolysis - Computing deposited mass, gas volume, charge, current, time, and Faraday's-law quantities.

  Electrochemical thermodynamics - Relating cell potential to Gibbs free energy, equilibrium constants, and spontaneity.
- **Batteries, corrosion, and applied cells** - Working with battery chemistry, corrosion processes, fuel cells, and applied electrochemical systems.

Organic Chemistry

Problems involving carbon-containing compounds, functional groups, reactions, mechanisms, stereochemistry, and structure determination. Avoiding and excluding open-ended retrosynthesis or mechanistic essays unless the answer is a directly checkable product, reagent, structure, label, or ordered sequence.

- **Structure and nomenclature** - Naming, drawing, and interpreting alkanes, alkenes, alkynes, aromatics, functional groups, substituents, and IUPAC structures.
- **Functional group chemistry** - Identifying and predicting behavior of alcohols, ethers, aldehydes, ketones, carboxylic acids, amines, amides, esters, and related groups.
- **Isomerism** - Identifying constitutional isomers, stereoisomers, conformers, enantiomers, diastereomers, meso compounds, and configurational relationships.
- **Stereochemistry** - Assigning R/S, E/Z, cis/trans, optical activity, chirality, and stereochemical outcomes.
- **Reaction mechanisms** - Identifying nucleophiles, electrophiles, intermediates, arrows, elementary steps, and mechanism classes.

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- **Substitution and elimination reactions** - Working with SN1, SN2, E1, E2, regioselectivity, stereochemistry, and competing pathways.
- **Addition and aromatic reactions** - Predicting alkene, alkyne, carbonyl, conjugate, electrophilic aromatic substitution, and nucleophilic aromatic substitution outcomes.
- **Carbonyl and carboxylic acid derivative chemistry** - Predicting reactions involving aldehydes, ketones, acids, acyl derivatives, enolates, and condensations.
- **Spectroscopic structure determination** - Interpreting IR, NMR, mass spectrometry, and UV-vis data to identify organic structures.
- **Synthesis and retrosynthesis** - Selecting reagents, products, intermediates, protecting groups, and feasible synthetic sequences.

Inorganic and Coordination Chemistry

Problems involving main-group chemistry, transition metals, coordination compounds, organometallics, and inorganic structure-reactivity relationships. Avoiding and excluding organic-specific mechanisms or materials-only problems unless the primary concept is inorganic bonding, structure, or reactivity.

Main-group chemistry - Working with periodic patterns, bonding, oxidation states, reactivity, and representative element compounds.

Transition metal chemistry - Computing oxidation states, d-electron counts, coordination numbers, and metal-centered properties.

Coordination compounds - Naming, formula writing, ligand counting, isomerism, geometry, and coordination-sphere structure.

Crystal field and ligand field theory - Determining splitting patterns, spin states, magnetic behavior, CFSE, and ligand-field effects.

Organometallic chemistry - Working with hapticity, electron counting, oxidative addition, reductive elimination, insertion, and ligand substitution.

Bioinorganic chemistry - Analyzing metal ions in biological systems, metalloproteins, cofactors, and metal-centered reactivity.

- **Solid-state inorganic chemistry** - Working with lattice structures, defects, band behavior, and inorganic extended solids.
- **Inorganic reaction mechanisms** - Classifying ligand substitution, electron transfer, inner-sphere and outer-sphere mechanisms, and trans effects.

Analytical Chemistry

Problems involving chemical measurement, quantitative analysis, instrumental methods, calibration, and uncertainty. Avoiding and excluding open-ended experimental critique unless

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the final answer is a calculation, classification, selected method, parameter value, or directly checkable interpretation.

- **Quantitative analysis** - Computing analyte amount, concentration, percent composition, purity, recovery, and dilution-corrected values.
- **Calibration and standardization** - Working with calibration curves, external standards, internal standards, standard additions, and linear response.
- **Titrimetric analysis** - Computing endpoint and equivalence quantities for acid-base, redox, complexometric, and precipitation titrations.
- **Gravimetric analysis** - Computing mass relationships, precipitate composition, analyte mass, and gravimetric factors.
- **Spectroscopic analysis** - Applying absorbance, Beer-Lambert law, transmittance, concentration determination, and spectral interpretation.
- **Chromatography** - Computing retention time, retention factor, resolution, plate number, selectivity, and separation efficiency.
- **Electroanalytical methods** - Working with potentiometry, voltammetry, conductometry, ion-selective electrodes, and electrochemical signals.
- **Error, uncertainty, and statistics** - Computing measurement uncertainty, confidence intervals, detection limits, propagation of error, precision, accuracy, and significant results.
- **Sampling and sample preparation** - Computing dilution factors, extraction recoveries, matrix effects, and sample preparation quantities.

Physical Chemistry and Spectroscopy

Problems involving molecular-level physical behavior, quantum chemistry, spectroscopy, statistical mechanics, and molecular thermodynamics. Avoiding and excluding purely general-chemistry thermodynamics or kinetics unless the primary task requires physical-chemistry models or molecular interpretation.

- **Quantum chemistry** - Working with particle-in-a-box models, operators, wavefunctions, energy levels, expectation values, and selection rules.

  Molecular orbital and electronic structure - Computing orbital occupancies, term symbols, electronic transitions, and qualitative electronic structure.

  Rotational and vibrational spectroscopy - Computing rotational constants, vibrational frequencies, energy spacings, and spectral transitions.

  Electronic spectroscopy - Working with UV-vis transitions, fluorescence, phosphorescence, selection rules, and electronic-state behavior.

  Magnetic resonance spectroscopy - Interpreting NMR, EPR, chemical shifts, coupling, multiplicity, and spin behavior.

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- **Statistical mechanics** - Computing partition functions, populations, Boltzmann factors, thermodynamic functions, and ensemble quantities.
- **Molecular thermodynamics** - Relating microscopic states to entropy, heat capacity, free energy, and equilibrium behavior.
- **Surface and interfacial chemistry** - Working with adsorption isotherms, surface tension, surface area, catalysis surfaces, and interfacial phenomena.

Biochemistry and Chemical Biology

Problems involving chemical processes in biological systems, biomolecules, enzyme behavior, metabolism, and biochemical thermodynamics. Avoiding and excluding broad biological explanation unless the primary task is chemical structure, reaction, binding, energetics, or quantitative biochemical analysis.

- **Biomolecular structure** - Identifying amino acids, peptides, proteins, nucleic acids, carbohydrates, lipids, cofactors, and functional groups.
- **Protein chemistry** - Working with pI, charge state, folding interactions, secondary structure, binding sites, and amino-acid properties.
- **Enzyme kinetics** - Computing Michaelis-Menten quantities, inhibition type, catalytic efficiency, turnover number, and rate behavior.
- **Biochemical thermodynamics** - Computing binding free energies, equilibrium constants, redox potentials, ATP coupling, and reaction favorability.
- **Metabolic pathways** - Identifying pathway intermediates, stoichiometry, yields, redox carriers, and pathway regulation points.
- **Acid-base behavior in biomolecules** - Computing protonation states, pH-dependent charge, buffer effects, and titration behavior of biomolecules.
- **Ligand binding and cooperativity** - Working with dissociation constants, fractional occupancy, Hill coefficients, and binding curves.
- **Chemical biology methods** - Interpreting labeling, probes, assays, sequencing chemistry, and chemically controlled biological measurements.

Materials, Polymer, and Solid-State Chemistry

Problems involving chemical structure and properties of extended materials, polymers, solids, nanomaterials, and functional materials. Avoiding and excluding mechanical engineering or device-design discussion unless the primary task is chemical composition, structure, property, or characterization.

Polymer chemistry - Computing degree of polymerization, molecular weight, repeat units, tacticity, copolymer composition, and polymerization stoichiometry.

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  Polymer properties - Working with glass transition, crystallinity, crosslinking, elasticity, solubility, and structure-property relationships.

  Solid-state structure - Analyzing lattice types, unit cells, packing, density, defects, vacancies, and interstitials.

  Electronic materials - Working with band gaps, doping, semiconductors, conductivity, and electronic structure of solids.

  Nanomaterials - Computing surface-area-to-volume ratios, particle size, quantum confinement, and nanoscale composition.

  Materials characterization - Interpreting XRD, microscopy, thermal analysis, spectroscopy, and diffraction-derived quantities.
- **Ceramic and ionic materials** - Working with ionic lattices, defects, conductivity, phase stability, and solid electrolytes.
- **Functional and soft materials** - Classifying or computing properties of gels, liquid crystals, supramolecular materials, and responsive materials.

Computational and Theoretical Chemistry

Problems involving computational models, molecular simulation, electronic structure methods, and theoretical chemical prediction. Avoiding and excluding open-ended software-use discussion unless the expected answer is a value, classification, selected method, parameter, or directly checkable computational result.

- **Electronic structure methods** - Classifying or applying Hartree-Fock, density functional theory, post-HF methods, basis sets, and correlation models.
- **Molecular mechanics** - Working with force fields, bonded and nonbonded terms, conformational energy, and parameterized molecular models.
- **Molecular dynamics** - Computing or interpreting trajectories, time steps, ensembles, temperature control, and molecular motion.
- **Monte Carlo methods** - Working with sampling, acceptance probabilities, ensembles, and stochastic molecular configurations.
- **Potential energy surfaces** - Identifying minima, transition states, reaction coordinates, saddle points, and energy barriers.
- **Computational thermochemistry** - Computing reaction energies, free energies, zero-point corrections, and thermochemical cycles.
- **Cheminformatics** - Working with molecular descriptors, fingerprints, similarity metrics, SMILES strings, and structure-property prediction.
- **Theoretical model selection** - Selecting appropriate computational models, approximations, or levels of theory for specified chemical tasks.

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Physics

Measurement, Units, and Experimental Physics

Problems involving physical quantities, units, measurement, uncertainty, dimensional analysis, and experimental interpretation. Avoiding and excluding advanced theoretical modeling unless the primary task is measurement, calibration, data interpretation, or unit-based reasoning.

  Units and dimensional analysis - Working with SI units, unit conversions, dimensions, dimensional consistency, scaling, and physical units.

  Significant figures and uncertainty - Computing significant figures, absolute uncertainty, relative uncertainty, percent uncertainty, and propagated uncertainty.

  Experimental design and controls - Identifying variables, controls, measurement procedures, calibration steps, and sources of error.

  Data analysis and fitting - Interpreting tables, plots, slopes, intercepts, regression fits, residuals, and extracted physical parameters.

- **Instrumentation and sensors** - Working with measurement devices, resolution, sensitivity, calibration curves, response functions, and instrumental limits.
- **Error analysis** - Computing random error, systematic error, precision, accuracy, standard deviation, standard error, and confidence intervals.
- **Physical estimation** - Performing order-of-magnitude estimates, Fermi estimates, scaling estimates, and plausibility checks.
- **Model validation** - Comparing measured data to physical models, expected trends, limiting cases, and experimental constraints.

Kinematics and Dynamics

Problems involving motion, forces, acceleration, Newton's laws, and the dynamics of particles. Avoiding and excluding energy-focused, rotational, relativistic, or field-theoretic problems unless the primary task is force-motion analysis.

- **One-dimensional kinematics** - Computing displacement, velocity, acceleration, time, and constant-acceleration motion in one dimension.
- **Two- and three-dimensional kinematics** - Working with projectile motion, relative motion, vector components, trajectories, and parametric motion.

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- **Newton's laws** - Applying force balance, net force, acceleration, free-body diagrams, and inertial or non-inertial frames.
- **Friction and contact forces** - Computing static friction, kinetic friction, normal forces, tension, drag, and contact interactions.
- **Circular motion** - Working with centripetal acceleration, centripetal force, tangential speed, angular speed, and radial dynamics.
- **Constraint dynamics** - Analyzing pulleys, inclined planes, connected bodies, Atwood machines, and constrained motion.
- **Drag and resistive forces** - Computing motion under linear drag, quadratic drag, terminal velocity, and velocity-dependent forces.
- **Non-inertial frames** - Working with apparent forces, accelerating frames, rotating frames, centrifugal effects, and Coriolis effects.

Energy, Momentum, and Rotational Mechanics

Problems involving conservation laws, work, energy, impulse, momentum, collisions, rotation, torque, and angular momentum. Avoiding and excluding purely force-motion problems unless conservation or rotational reasoning is central.

Work and kinetic energy - Computing work, kinetic energy, work-energy relationships, and variable-force work.

Potential energy and conservative forces - Working with gravitational, elastic, electric, and general potential energy functions.

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  Conservation of mechanical energy - Solving problems using energy conservation, energy conversion, and energy diagrams.

  Power and efficiency - Computing power, rate of energy transfer, mechanical efficiency, and dissipated energy.

  Linear momentum and impulse - Working with momentum, impulse, force-time relationships, and momentum conservation.

  Collisions and explosions - Classifying and computing elastic, inelastic, perfectly inelastic, and multi-body collision outcomes.
- **Rotational kinematics and dynamics** - Computing angular displacement, angular velocity, angular acceleration, torque, and rotational motion.
- **Moment of inertia** - Working with rotational inertia, parallel-axis theorem, composite bodies, and mass distributions.
- **Angular momentum** - Computing angular momentum, torque-angular momentum relationships, conservation, and rotational collision outcomes.
- **Static equilibrium** - Solving torque balance, force balance, center of mass, stability, and rigid-body equilibrium problems.

Oscillations, Waves, and Acoustics

Problems involving periodic motion, mechanical waves, sound, superposition, resonance, and wave phenomena. Avoiding and excluding quantum wave mechanics or electromagnetic optics unless the primary task is classical oscillation or wave behavior.

- **Simple harmonic motion** - Computing period, frequency, amplitude, phase, displacement, velocity, acceleration, and energy in oscillators.
- **Damped and driven oscillations** - Working with damping, resonance, quality factor, forced oscillations, and steady-state response.
- **Mechanical waves** - Computing wave speed, wavelength, frequency, period, amplitude, phase, and wave equations.
- **Superposition and interference** - Analyzing constructive interference, destructive interference, beats, standing waves, and phase differences.
- **Sound waves** - Working with intensity, decibels, Doppler effect, harmonics, resonance tubes, and acoustic standing waves.
- **Wave propagation** - Computing wavefront behavior, boundary reflection, transmission, dispersion, and group velocity.
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- **Fourier and normal modes** - Identifying mode shapes, harmonic content, frequency spectra, and normal-mode frequencies.
- **Nonlinear waves** - Working with solitons, shock waves, nonlinear oscillators, and amplitude-dependent wave behavior.

Thermodynamics and Statistical Physics

Problems involving heat, temperature, work, entropy, engines, thermal processes, and statistical descriptions of matter. Avoiding and excluding chemical thermodynamics or fluid mechanics unless the primary task is physical thermodynamics or statistical behavior.

Temperature and heat - Computing heat transfer, temperature change, heat capacity, specific heat, and calorimetry quantities.

Thermal expansion - Working with linear, area, and volume expansion and temperature-dependent size changes.

Ideal gases - Applying ideal gas relationships, kinetic theory, molecular speeds, pressure, volume, temperature, and moles.

Thermodynamic processes - Computing work, heat, internal energy, and state changes for isothermal, adiabatic, isobaric, and isochoric processes.

First law of thermodynamics - Applying energy conservation to thermodynamic systems, cycles, and processes.

Entropy and second law - Computing entropy changes, irreversibility, heat flow direction, and thermodynamic feasibility.

- **Heat engines and refrigerators** - Computing efficiency, coefficient of performance, Carnot limits, and cyclic process quantities.
- **Statistical mechanics** - Working with microstates, macrostates, partition functions, Boltzmann factors, ensembles, and equilibrium distributions.

Electricity and Magnetism

Problems involving electric charges, fields, potentials, currents, magnetic fields, induction, and Maxwellian electromagnetic behavior. Avoiding and excluding circuit-only problems unless the primary task is electromagnetic field or force analysis.

- **Electrostatics** - Computing electric forces, electric fields, charge distributions, Coulomb's law, and superposition.
- **Electric potential** - Working with voltage, potential energy, equipotentials, point charges, conductors, and potential differences.
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- **Gauss's law** - Applying flux and symmetry to electric fields from spherical, cylindrical, planar, or conducting charge distributions.
- **Capacitance and dielectrics** - Computing capacitance, stored energy, dielectric effects, charge, voltage, and equivalent capacitance.
- **Electric current and resistance** - Working with current density, conductivity, resistivity, drift velocity, and microscopic current models.
- **Magnetic forces and fields** - Computing magnetic forces on charges or wires, magnetic fields from currents, and charged-particle motion.
- **Ampère's law and Biot-Savart law** - Computing magnetic fields from wires, loops, solenoids, and symmetric current distributions.
- **Electromagnetic induction** - Applying Faraday's law, Lenz's law, induced emf, motional emf, and time-varying magnetic flux.
- **Maxwell's equations** - Working with integral or differential forms, displacement current, electromagnetic waves, and field relationships.

Circuits and Electronics

Problems involving circuit elements, circuit laws, transient behavior, AC circuits, and electronic devices. Avoiding and excluding field-theoretic electromagnetism unless the primary task is circuit analysis or electronic behavior.

- **DC circuits** - Applying Ohm's law, Kirchhoff's laws, series and parallel resistors, node voltages, and loop currents.

  RC, RL, and RLC circuits - Computing transient charging, discharging, time constants, oscillations, damping, and steady-state behavior.

  AC circuits - Working with impedance, reactance, phasors, RMS quantities, resonance, phase shifts, and AC power.

  Circuit power and energy - Computing electrical power, dissipated energy, stored energy, efficiency, and power transfer.

  Capacitor and inductor networks - Computing equivalent capacitance, equivalent inductance, stored energy, and circuit response.

  Diodes and semiconductor devices - Working with diode behavior, rectification, threshold voltage, simple transistor behavior, and device characteristics.

  Operational amplifier circuits - Computing gains, outputs, virtual ground behavior, filters, and ideal op-amp circuit quantities.

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- **Signal filtering and frequency response** - Analyzing low-pass, high-pass, band-pass, cutoff frequency, gain, and transfer functions.

Optics

Problems involving light propagation, reflection, refraction, lenses, mirrors, interference, diffraction, and optical instruments. Avoiding and excluding quantum optics or electromagnetic-wave derivations unless the primary task is optical calculation or classification.

- **Geometric optics** - Computing ray paths, image distance, object distance, magnification, focal length, and image orientation.
- **Reflection and mirrors** - Applying reflection law, mirror equation, spherical mirrors, plane mirrors, and image formation.
- **Refraction and lenses** - Applying Snell's law, critical angle, total internal reflection, thin lenses, and lens combinations.
- **Optical instruments** - Working with microscopes, telescopes, cameras, magnifiers, resolving power, and image formation systems.
- **Interference** - Computing path differences, phase differences, fringe spacing, thin-film interference, and double-slit patterns.
- **Diffraction** - Working with single-slit diffraction, diffraction gratings, angular resolution, and diffraction-limited behavior.
- **Polarization** - Analyzing polarized light, Malus's law, Brewster's angle, polarizers, and polarization states.
- **Wave optics and coherence** - Working with coherence length, wavefronts, optical path length, and wave-based light behavior.

Fluid Mechanics and Continuum Physics

Problems involving fluids, pressure, buoyancy, flow, viscosity, and continuum behavior of matter. Avoiding and excluding thermodynamics or materials science unless the primary task is fluid or continuum calculation.

- **Fluid statics** - Computing pressure, hydrostatic force, density effects, manometers, and pressure-depth relationships.
- **Buoyancy** - Applying Archimedes' principle, floating conditions, apparent weight, displaced volume, and stability.
- **Fluid flow** - Computing flow rate, continuity, velocity fields, and incompressible flow relationships.
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- **Bernoulli's principle** - Applying pressure-speed-height relationships, Venturi effects, efflux speed, and energy conservation in fluids.

  Viscosity and drag - Working with viscous flow, Poiseuille's law, Reynolds number, laminar flow, turbulent flow, and drag forces.

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  Surface tension and capillarity - Computing surface forces, capillary rise, pressure differences, and interfacial effects.

  Elasticity and stress-strain - Working with Young's modulus, shear modulus, bulk modulus, strain, stress, and elastic deformation.

  Continuum waves and fields - Analyzing waves, stresses, strains, and field quantities in continuous media.

Relativity

Problems involving special relativity, spacetime, relativistic energy and momentum, and gravitational relativistic effects. Avoiding and excluding philosophical interpretation or tensor-field derivations unless the expected answer is a directly checkable value, classification, or expression.

- **Lorentz transformations** - Computing transformed positions, times, velocities, and spacetime intervals between inertial frames.
- **Time dilation and length contraction** - Working with proper time, proper length, moving clocks, moving rods, and relativistic factors.
- **Relativistic velocity addition** - Computing combined velocities and frame-dependent relative speeds.
- **Relativistic momentum and energy** - Computing rest energy, total energy, kinetic energy, momentum, and mass-energy relationships.
- **Spacetime intervals and causality** - Classifying intervals as timelike, spacelike, or lightlike and determining causal relationships.
- **Relativistic Doppler effect** - Computing observed frequency, wavelength shifts, and source-observer motion effects.
- **Equivalence principle and gravitational time dilation** - Working with weak-field gravitational effects, clocks, acceleration, and redshift.
- **General relativity basics** - Computing or identifying geodesic behavior, Schwarzschild radius, gravitational lensing quantities, and simple spacetime effects.

Quantum Mechanics

Problems involving wavefunctions, operators, quantization, measurement, spin, and quantum systems. Avoiding and excluding proof-based formalism or interpretive essays unless the answer is a directly checkable value, state, probability, eigenvalue, or classification.

- **Wavefunctions and probability** - Computing normalization, probability density, expectation values, and measurement probabilities.
- **Schrödinger equation systems** - Working with particles in boxes, harmonic oscillators, barriers, wells, and simple potentials.
- **Operators and observables** - Computing eigenvalues, eigenfunctions, commutators, expectation values, and uncertainty relationships.

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- **Angular momentum and spin** - Working with spin states, angular momentum quantum numbers, addition rules, and measurement outcomes.
- **Quantum tunneling** - Computing barrier transmission, qualitative tunneling behavior, and exponential attenuation.
- **Perturbation theory** - Computing first-order energy shifts, approximate eigenvalues, and perturbative corrections.

  Identical particles - Working with fermions, bosons, Pauli exclusion, exchange symmetry, and occupation behavior.

  Quantum information basics - Computing qubit states, measurement outcomes, simple gates, entanglement labels, and state transformations.

Atomic, Molecular, and Optical Physics

Problems involving atoms, molecules, spectra, transitions, lasers, and light-matter interaction. Avoiding and excluding chemistry-focused molecular structure unless the primary task is physical spectra, quantum states, or radiative behavior.

- **Atomic energy levels** - Computing transition energies, wavelengths, frequencies, ionization energies, and quantum numbers.
- **Hydrogenic atoms** - Working with Bohr model quantities, Rydberg formula, energy levels, radii, and spectral lines.
- **Fine structure and hyperfine structure** - Identifying energy splittings, selection rules, spin-orbit effects, and transition patterns.
- **Molecular rotation and vibration** - Computing rotational energies, vibrational energies, transition frequencies, and molecular spectra.
- **Selection rules** - Identifying allowed and forbidden transitions using angular momentum, parity, and quantum-number changes.
- **Lasers and optical pumping** - Working with population inversion, stimulated emission, gain, thresholds, and cavity behavior.
- **Light-matter interaction** - Computing absorption, emission, scattering, cross sections, and transition probabilities.
- **Spectroscopic analysis** - Interpreting spectra, line broadening, Doppler broadening, lifetime broadening, and spectral resolution.

Nuclear and Particle Physics

Problems involving nuclei, radioactive decay, nuclear reactions, elementary particles, conservation laws, and high-energy interactions. Avoiding and excluding open-ended model discussion unless the expected answer is a value, particle, reaction, conservation classification, or directly checkable output.

- **Nuclear structure** - Working with protons, neutrons, isotopes, binding energy, mass defect, nuclear radius, and nuclear stability.
- **Radioactive decay** - Computing half-life, activity, decay constants, remaining nuclei, decay chains, and decay modes.
- **Nuclear reactions** - Balancing nuclear equations, computing Q-values, reaction thresholds, and emitted particles.

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- **Fission and fusion** - Computing energy release, reaction products, mass-energy conversion, and chain-reaction quantities.
- **Radiation and matter** - Working with attenuation, dose, shielding, stopping power, and radiation interactions.
- **Particle classification** - Identifying leptons, quarks, hadrons, baryons, mesons, bosons, antiparticles, and quantum numbers.
- **Conservation laws** - Applying charge, baryon number, lepton number, energy, momentum, spin, and strangeness conservation.

  Standard Model interactions - Classifying strong, weak, electromagnetic, and gravitational interactions and associated mediator particles.

Condensed Matter and Materials Physics

Problems involving solids, electronic properties, lattice structure, phonons, magnetism, superconductivity, and material behavior. Avoiding and excluding chemistry-only composition questions unless the primary task is physical structure or material property analysis.

- **Crystal structure** - Working with lattices, unit cells, Bravais lattices, Miller indices, packing, reciprocal lattices, and diffraction.
- **Band theory** - Computing or identifying band gaps, Fermi levels, density of states, conductors, insulators, and semiconductors.
- **Semiconductors** - Working with doping, carrier concentrations, p-n junctions, depletion regions, mobility, and conductivity.
- **Phonons and lattice vibrations** - Computing vibrational modes, heat capacity, dispersion, and thermal transport quantities.
- **Magnetism** - Classifying diamagnetism, paramagnetism, ferromagnetism, antiferromagnetism, domains, and magnetic response.
- **Superconductivity** - Working with critical temperature, Meissner effect, Cooper pairs, critical fields, and flux quantization.
- **Thermal and electrical transport** - Computing conductivity, resistivity, mobility, diffusion, thermal conductivity, and Hall effect quantities.
- **Soft matter and polymers** - Analyzing elastic response, polymer scaling, liquid crystals, colloids, and mesoscale material behavior.

Astrophysics, Cosmology, and Gravitation

Problems involving celestial motion, stars, galaxies, cosmology, gravitation, and astronomical observation. Avoiding and excluding geology or atmospheric science unless the primary task is astrophysical or gravitational reasoning.

- **Newtonian gravitation** - Computing gravitational force, gravitational field, orbital speed, escape speed, and gravitational potential energy.
- **Orbital mechanics** - Working with Kepler's laws, orbital periods, semi-major axes, eccentricity, satellites, and two-body systems.

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- **Stellar properties** - Computing luminosity, temperature, radius, magnitude, spectra, stellar classification, and blackbody behavior.
- **Stellar evolution** - Identifying main sequence behavior, red giants, white dwarfs, neutron stars, black holes, and evolutionary endpoints.
- **Galaxies and large-scale structure** - Working with galaxy classification, rotation curves, dark matter evidence, redshift, and distance measures.
- **Cosmology** - Computing Hubble-law quantities, scale factor relationships, cosmic expansion, density parameters, and cosmic microwave background quantities.
- **Observational astronomy** - Interpreting parallax, magnitude systems, spectra, Doppler shifts, telescopes, and detector data.
- **Compact objects and relativistic astrophysics** - Working with Schwarzschild radius, accretion, pulsars, gravitational waves, and relativistic orbital effects.

Plasma Physics and Electromagnetic Media

Problems involving ionized gases, charged-particle collective behavior, waves in media, and electromagnetic response of materials. Avoiding and excluding circuit-only or simple gas-law problems unless the primary task is plasma or medium behavior.

- **Plasma parameters** - Computing plasma frequency, Debye length, Debye shielding, temperature, density, and ionization fraction.
- **Charged-particle motion** - Working with cyclotron motion, gyrofrequency, Larmor radius, magnetic mirrors, and drift motion.
- **Magnetohydrodynamics** - Applying fluid-like plasma models, frozen-in fields, Alfvén waves, and magnetic pressure.
- **Plasma waves and instabilities** - Identifying wave modes, dispersion relations, growth rates, and instability conditions.
- **Electromagnetic waves in media** - Computing phase velocity, refractive index, attenuation, impedance, and dispersion in materials.
- **Dielectric and magnetic media** - Working with permittivity, permeability, polarization, magnetization, susceptibility, and constitutive relationships.
- **Waveguides and cavities** - Computing modes, cutoff frequencies, resonances, and standing electromagnetic waves.
- **Space and fusion plasmas** - Working with solar wind, magnetospheres, tokamaks, confinement, and plasma heating quantities.

Example Problems

Example Problems

[Problem #1](#)
[Problem #2](#)
[Problem #3](#)
[Problem #4](#)
[Problem #5](#)

[Problem #6](#)

⚠**Please note:** problems must adhere to the LaTeX Standards of this project laid out in the Riga Answer Formatting Guide. This section is simply meant to provide visibility on previously accepted problems to reference and use as inspiration, and are not representative of the formatting guidance that is expected on Riga.

Example Problem #1

Field: Chemistry

Domain: Electrochemistry **Sub-Domain:**

Nernst equation

PROBLEM PROMPT:

A one-electron redox couple is described by the electrode reaction

$$O + e^- \rightleftharpoons R,$$

with standard reduction potential E° for the free species O and R. The solution is ideal and dilute, so activities are given by concentration divided by the standard concentration c°.
Only the oxidized form binds a ligand L:

$$O + L \rightleftharpoons OL,$$

with dimensionless formation constant

$$\beta^\circ = \frac{a_{OL}}{a_O a_L}.$$

The total analytical metal concentration is

$$C_M = [R] + [O] + [OL],$$

and the total analytical ligand concentration is

$$C_L = [L] + [OL].$$

The system is titrated electrochemically until the analytical half-reduction point is reached, meaning

$$[R] = [O] + [OL] = \frac{C_M}{2}.$$

Assume that complexation is always at equilibrium and that only free O and free R enter the Nernst equation.
Determine the electrode potential $E_{1/2,\mathrm{app}}$ at this analytical half-reduction point in terms of E°, R, T, F, C_M, C_L, β°, and c°.
Return only the final expression.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Write the Nernst equation for the free redox pair
For the reduction reaction

$$O + e^- \rightleftharpoons R$$

the Nernst equation is

$$E = E^\circ + \frac{RT}{F} \ln\left(\frac{a_O}{a_R}\right)$$

Because the solution is ideal and dilute,

$$a_O = \frac{[O]}{c^\circ}, \qquad a_R = \frac{[R]}{c^\circ}$$

Therefore we get

$$E = E^\circ + \frac{RT}{F} \ln\left(\frac{[O]}{[R]}\right)$$

Step 2: Express the complexation equilibrium in concentration form
The ligand binding equilibrium is:

$$O + L \rightleftharpoons OL$$

The dimensionless formation constant is:

$$\beta^\circ = \frac{a_{OL}}{a_O a_L}.$$

Using

$$a_{OL} = \frac{[OL]}{c^\circ}, \qquad a_O = \frac{[O]}{c^\circ}, \qquad a_L = \frac{[L]}{c^\circ}$$

we get

$$\beta^\circ = \frac{[OL]/c^\circ}{([O]/c^\circ)([L]/c^\circ)}.$$

Therefore

$$\beta^\circ = \frac{[OL]c^\circ}{[O][L]}$$

So now

$$[OL] = \frac{\beta^\circ}{c^\circ}[O][L]$$

If we define

$$K = \frac{\beta^\circ}{c^\circ}$$

Then

$$[OL] = K[O][L]$$

Step 3: We now use the analytical half-reduction condition
At the analytical half-reduction point

$$[R] = [O] + [OL] = \frac{C_M}{2}$$

Let

$$X = \frac{C_M}{2}$$

Then

$$[R] = X$$

and

$$[O] + [OL] = X$$

Using

$$[OL] = K[O][L]$$

we get

$$[O] + K[O][L] = X$$

Therefore,

$$[O](1 + K[L]) = X$$

So now

$$[O] = \frac{X}{1 + K[L]}$$

Step 4: Now we can substitute into the Nernst equation
Since

$$[R] = X$$

and

$$[O] = \frac{X}{1 + K[L]}$$

we have

$$\frac{[O]}{[R]} = \frac{X/(1 + K[L])}{X}$$

Therefore,

$$\frac{[O]}{[R]} = \frac{1}{1 + K[L]}$$

So the electrode potential becomes

$$E_{1/2,\text{app}} = E^\circ + \frac{RT}{F} \ln\left(\frac{1}{1 + K[L]}\right)$$

Thus,

$$E_{1/2,\text{app}} = E^\circ - \frac{RT}{F} \ln\left(1 + K[L]\right)$$

Step 5: Next we solve for the free ligand concentration
The total ligand concentration is

$$C_L = [L] + [OL]$$

At the half-reduction point,

$$[O] + [OL] = X$$

And

$$[O] = \frac{X}{1 + K[L]}$$

Therefore,

$$[OL] = X - [O]$$

We substitute the expression for $[O]$:

$$[OL] - X - \frac{X}{1 + K[L]}$$

So

$$[OL] - X\left(1 - \frac{1}{1 + K[L]}\right)$$

Therefore

$$[OL] = X\left(\begin{matrix} K[L] \\ 1 + K[L] \end{matrix}\right)$$

Now we substitute into the ligand mass balance

$$C_L = [L] + X\left(\frac{K[L]}{1 + K[L]}\right)$$

If we let

$$y = 1 + K[L]$$

Then

$$[L] = \begin{matrix} y - 1 \\ K \end{matrix}$$

The ligand balance now becomes

$$C_L = \frac{y - 1}{K} + X\left(\frac{y - 1}{y}\right)$$

Multiply both sides by Ky:

$$KC_L y = y(y - 1) - KX(y - 1)$$

Expand

$$KC_L y = y^2 - y + KXy - KX.$$

Then bring all terms to one side

$$0 = y^2 + (KX - 1 - KC_L)\, y - KX$$

So y satisfies

$$y^2 + (KX - 1 - KC_L)\, y - KX - 0$$

The physical root is the positive root:

#2

Field: Physics

Domain: Quantum Mechanics **Sub-Domain:**

Perturbation theory **PROBLEM PROMPT:**

Example Problem

Consider the Hamiltonian of a single unperturbed quantum harmonic oscillator, expressed using the typical bosonic ladder operators.
$H_0 = \hbar\omega(a^\dagger a + \frac{1}{2})$
A perturbation is applied to the aforementioned system, such that the oscillator's mass m and elasticity k become some new m' and k', but its angular frequency ω remains constant. An additive, normal-ordered perturbation V is said to describe such a change.
$H = H_0 + V$
Find an expression for the diagonal elements $\langle z|V|z\rangle$, where $z \geq 1$.

STEP-BY-STEP SOLUTION & ANSWER:

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p^2 and q^2 as

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and factoring

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s in V become

Example Problem #3

Field: Chemistry

Domain: Organic Chemistry

Sub-Domain: Synthesis and retrosynthesis

PROBLEM PROMPT:

I am seeking advice on the structural assignment of the final product obtained from a multistep synthetic sequence involving an initially simple aliphatic ketone substrate. The study was designed to evaluate whether a functionalized intermediate derived from pentan-3-one could undergo a high-temperature reaction with an allylic alcohol derivative to furnish a rearranged or otherwise transformed product.

In the first stage, pentan-3-one was subjected to a four-step reaction sequence under the following conditions:

1. Treatment with formaldehyde, dimethylamine, and hydrochloric acid, represented as CH_2O, Me_2NH, and HCl. 2. Subsequent treatment with potassium carbonate, K_2CO_3. 3. Alkylation with methyl iodide, MeI. 4. Treatment with potassium hydroxide, KOH.

The product obtained from this sequence was then reacted with (2*Z*)-but-2-en-1-ol under high-temperature conditions at 290°C. After completion of the reaction, the crude material was subjected to an acidic aqueous work-up using dilute HCl. The organic product was then extracted into dichloromethane, CH_2Cl_2, and the organic phase was washed successively with aqueous sodium bicarbonate, $NaHCO_3$, and brine. The organic layer was dried over magnesium sulfate, $MgSO_4$, concentrated, and purified by column chromatography.

I would like assistance in assigning the structure of the ultimate isolated product from this sequence. Please identify the final product using its standard InChI string format.

STEP-BY-STEP SOLUTION & ANSWER:

d in situ from
a $-CH_2NMe_2$

s the dimethy-
a good leaving

a β-proton is
thylpent-1-en-

electron-poor
ydroxymethyl-

hermal $[4 + 2]$
lcohol-bearing
bearing alkene
6 double bond.
4-dihydro-2H-
hile, combined
is later carried

ctron-rich enol
s stabilized by

carbenium ion
ethyl ketone
pen keto-diol
$_2$OH group is

one is reproto-
imary CH_2OH
ecomes a ketal

-2, 4-dimethyl-
2C. The ketal
4, and the two
le aggregation

$)6 - 11 - 10/h7$

$6H2, 1 - 3H3$

#4

Field: Physics

Domain: Quantum Mechanics **Sub-Domain:** Identical particles

PROBLEM PROMPT:

Example Problem

Two identical photons occupy five optical modes labeled $1, 2, 3, 4, 5$. A Fock state is written by listing the occupied modes. For example, 14 means one photon in mode 1 and one photon in mode 4.

The initial state is a classical mixture over the ten states $12, 13, 14, 15, 23, 24, 25, 34, 35, 45$. If the initial state is ij, its probability is proportional to $\lambda_i \lambda_j$, where $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$, $\lambda_4 = 5$, and $\lambda_5 = 8$.

After the initial state is drawn, one unordered pair of modes is chosen but not revealed. Pair ij is chosen with probability proportional to $\mu_i \mu_j$, where $\mu_1 = 1$, $\mu_2 = 3$, $\mu_3 = 4$, $\mu_4 = 7$, and $\mu_5 = 11$. A balanced beam splitter is applied to the chosen pair of modes. For a beam splitter on modes i and j, use the transformation $a_i^\dagger \mapsto (a_i^\dagger + a_j^\dagger)/\sqrt{2}$ and $a_j^\dagger \mapsto (a_i^\dagger - a_j^\dagger)/\sqrt{2}$.

Then a second unordered pair of modes is chosen uniformly at random from the six pairs that share exactly one mode with the first chosen pair. This second pair is also not revealed. A balanced beam splitter with the same transformation rule is applied to the second chosen pair.

After the two beam splitters, the photon number in every mode is measured. The outcome is one photon in mode 4, one photon in mode 5, and no photons in modes $1, 2, 3$.

Given this measurement outcome, what is the exact probability that the initial state was 45? Give the answer as a simplified exact fraction.

STEP-BY-STEP SOLUTION & ANSWER:

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state is ij and

pair is chosen
choices, define

. This scaling

ond pairs are

rst pair choices

Bayes' rule, so
o products are

Example Problem #5

Field: Biology

Domain: Genetics and Genomics **Sub-**

Domain: Linkage and recombination

PROBLEM PROMPT:

A researcher is examining the inheritance of a neurodevelopmental phenotype associated with the murine *Mid*1 locus. The genomic structure of this gene spans the pseudoautosomal boundary (PAB). Exons 1–3 are X-unique, and exons 4–10 are located in the pseudoautosomal region (PAR).
A recessive mutation at a site near the 3′ end of exon 3 results in cerebral hypoplasia. Severity is determined by the number of exons impacted by the mutation. The most severe phenotype occurs when 8 exons are affected, and the severity of the phenotype lessens as the number of impacted exons decreases.
In the parental generation, a wild-type female mouse is crossed with a male carrying a *Mid*1 +1 frameshift at the 3′ end of exon 3, affecting exons 3–10.
Three F_1 pups from this cross are mated to unrelated mice once they reach reproductive maturity.
An F_1 male is mated with a female heterozygous for the *Mid*1 +1 frameshift at the 3′ end of exon 3, affecting exons 3–10. They have 9 pups in their litter: 6 males and 3 females.
An F_1 female is mated with a wild-type male. They have 7 pups: 3 females and 4 males.
Another F_1 male is crossed with a female homozygous for the *Mid*1 +1 frameshift at the 3′ end of exon 3, affecting exons 3–10. They have 5 pups: 3 females and 2 males.
During male meiosis, the recombination frequency within the PAR segment of the *Mid*1 gene is 33.3%. During female meiosis, this recombination frequency is negligible.
Assuming a fully functional wild-type allele is completely dominant, and the most fully functional allele is dominant when no fully functional wild-type allele is present, how many of the pups described in the F_2 generation are expected to have severe cerebral hypoplasia? Provide your answer as a whole number using standard rounding rules if needed.

STEP-BY-STEP SOLUTION & ANSWER:

t at the 3' end
he most severe
rs in the PAR,
on 3, upstream
n 3 frameshift.
e X^m allele.
X^mY. All F1
. All F1 males

emale $XwtX^m$.
from the father
robability 1/2.
le pups inherit
minant, none of
). Cross 1 total

wild-type male
erit Y from the
robability 1/2.
ips inherit Xwt
dominant, none
2 are $3 \times 0 = 0$.

mutant female
erit Y from the
ed severe males
he homozygous
wild-type allele
everely affected

mber of severe

Example Problem #6

Field: Physics

Domain: Quantum Mechanics **Sub-Domain:**

Identical particles

PROBLEM PROMPT:

Two identical photons occupy six optical modes labeled $1, 2, 3, 4, 5, 6$. A Fock state is written by listing the occupied modes. For example, 26 means one photon in mode 2 and one photon in mode 6. The initial state is a classical mixture over the fifteen states $12, 13, 14, 15, 16, 23, 24, 25, 26, 34, 35, 36, 45, 46, 56$. If the initial state is ij, its probability is proportional to $\lambda_i \lambda_j$, where $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$, $\lambda_4 = 5$, $\lambda_5 = 7$, and $\lambda_6 = 11$.

After the initial state is drawn, a first unordered pair of modes is chosen but not revealed. Pair ij is chosen with probability proportional to $\mu_i \mu_j$, where $\mu_1 = 2$, $\mu_2 = 3$, $\mu_3 = 5$, $\mu_4 = 7$, $\mu_5 = 11$, and $\mu_6 = 13$. A balanced beam splitter is applied to that pair. For a beam splitter on pair ij with $i < j$, use the transformation $a_i^\dagger \mapsto (a_i^\dagger + a_j^\dagger)/\sqrt{2}$ and $a_j^\dagger \mapsto (a_i^\dagger - a_j^\dagger)/\sqrt{2}$.

Then a second unordered pair of modes is chosen uniformly at random from the six pairs that are disjoint from the first pair. This second pair is not revealed, and the same balanced beam splitter rule is applied.

Then a third unordered pair of modes is chosen uniformly at random from the four pairs that contain one mode from the first chosen pair and one mode from the second chosen pair. This third pair is also not revealed, and the same balanced beam splitter rule is applied.

After the three beam splitters, the photon number in every mode is measured. The outcome is one photon in mode 5, one photon in mode 6, and no photons in modes $1, 2, 3, 4$.

Given this measurement outcome, what is the exact probability that the initial state was 56? Give the answer as a simplified exact fraction.

STEP-BY-STEP SOLUTION & ANSWER:

factors for the
rd hidden pair
ently.
s in modes p, q,
e the pair, then

am splitter the

mplitude of the
$A_{ij}|^2$. To avoid

ible third pairs
l cancels in the

n pair 15. The

final state 56

r. To keep the
nes separately.

bution for first

4, 0.

120, 142296, 168

se first pair 15.

28.

$0 + 40656$.

od for that first
ates.

Step 7: Applying the same rule from Step 6 to each first pair gives the following check list. Each line gives the first pair pq, the value of $C_{56}(pq)$, the numerator value $N(pq) = 77C_{56}(pq)$, and the evidence value $E(pq)$.
For $pq = 12$, the values are

$$C_{56}(12) = 240, \qquad N(12) = 18480, \qquad E(12) = 25608.$$

For $pq = 13$, the values are

$$C_{56}(13) = 400, \qquad N(13) = 30800, \qquad E(13) = 42680.$$

For $pq = 14$, the values are

$$C_{56}(14) = 560, \qquad N(14) = 43120, \qquad E(14) = 59752.$$

For $pq = 15$, the values are

$$C_{56}(15) = 528, \qquad N(15) = 40656, \qquad E(15) = 62546.$$

For $pq = 16$, the values are

$$C_{56}(16) = 624, \qquad N(16) = 48048, \qquad E(16) = 68614.$$

For $pq = 23$, the values are

$$C_{56}(23) = 600, \qquad N(23) = 46200, \qquad E(23) = 64020.$$

For $pq = 24$, the values are

$$C_{56}(24) = 840, \qquad N(24) = 64680, \qquad E(24) = 89628.$$

For $pq = 25$, the values are

$$C_{56}(25) = 792, \qquad N(25) = 60984, \qquad E(25) = 101475.$$

For $pq = 26$, the values are

$$C_{56}(26) = 936, \qquad N(26) = 72072, \qquad E(26) = 109161.$$

For $pq = 34$, the values are

$$C_{56}(34) = 1400, \qquad N(34) = 107800, \qquad E(34) = 149380.$$

For $pq = 35$, the values are

$$C_{56}(35) = 1320, \qquad N(35) = 101640, \qquad E(35) = 181665.$$

For $pq = 36$, the values are

$$C_{56}(36) = 1560, \qquad N(36) = 120120, \qquad E(36) = 192075.$$

For $pq = 45$, the values are

$$C_{56}(45) = 1848, \qquad N(45) = 142296, \qquad E(45) = 288519.$$

For $pq = 46$, the values are

$$C_{56}(46) = 2184, \qquad N(46) = 168168, \qquad E(46) = 296205.$$

For $pq = 56$, the values are

$$C_{56}(56) = 0, \qquad N(56) = 0, \qquad E(56) = 169884.$$

For example, the fourth line has first pair 15. It matches Step 6 because $C_{56}(15) = 528$, so

Step 8: Add the numerator column from Step 7:

$$18480 + 30800 + 43120 + 40656 + 48048 + 46200 + 64680 + 60984 + 72072 + 107800$$

$$+101640 + 120120 + 142296 + 168168 + 0 = 1065064.$$

Add the evidence column from Step 7:

$$25608 + 42680 + 59752 + 62546 + 68614 + 64020 + 89628 + 101475 + 109161 + 149380$$

$$+181665 + 192075 + 288519 + 296205 + 169884 = 1901212.$$

Therefore the posterior probability before simplification is

$$\frac{1065064}{1901212}.$$

Step 9: Simplify the fraction:

$$\frac{1065064}{1901212} = \frac{266266}{475303}.$$

Therefore the exact probability is

$$\frac{266266}{475303}.$$

Final Answer: $\boxed{\frac{266266}{475303}}$

Final Answer:

$\frac{266266}{475303}$

Appendix LaTeX Formatting

Appendix: LaTeX Formatting

General Math Formatting Guidelines
- LaTeX Formatting
- Polynomial Formatting
- Fractional Expressions
- Units and Measurements
- Special Characters and Symbols
- Rounding and Approximations
- Uniformity

Examples of Accepted LaTeX Formatting

This document outlines acceptable LaTeX formatting requirements:

mhchem is now supported on the Snorkel platform! Currently `chemfig` is not supported

General Formatting Guidelines

Use LaTeX notation for mathematical symbols in prompts, final answers, and step rewrites. Avoid using LaTeX for non-mathematical formatting (i.e., bullet points, bold, etc.).

LaTeX Formatting

- $...$ and $$...$$ are allowed
- \(...\) and \[...\] are NOT allowed

Braces

- Avoid use of unnecessary braces that don't render (e.g., {\frac{3}{4}}\cdot{\frac{5}{8}})
- Make sure braces are escaped when they should be visible (e.g., \in\{1,2,3\}

Polynomial Formatting

- Always arrange polynomial terms in descending order of exponents

Fractional Expressions

- Use inline fractions (e.g., \frac{3}{4}).
- Simplify fractions whenever possible unless the unsimplified form adds context.

Units and Measurements

- Include units in annotations whenever relevant (e.g., 9.8 \, \text{m/s}^2.

Special Characters and Symbols

- Use proper mathematical symbols (e.g., \leq instead of <=).
- Define all special symbols used.

Other Style

- No use of \item[] for lists
- No use of **...** to bold Steps or Titles
- No LaTeX commands only valid in KaTeX (e.g. \lang instead of \langle)

Rounding and Approximations

If decimal approximations are required in the final answer, rounding instructions must be specified within the prompt (e.g., round to at least four decimal places). The model's rounding choices cannot be considered errors without these instructions.

Uniformity

Maintain consistent formatting for all submissions.

Examples of Accepted LaTeX Formatting

Category	*Numerical*	*LaTeX Equivalent*	*Rendered Form*
Fractions	*1/2*	*\frac{1}{2}*	$\frac{1}{2}$
	1 / (2+3)	*\frac{1}{2+3}*	$\frac{1}{2+3}$
Infinity	∞	\infty	∞
Exponents	*2^3*	*2^{3}*	2^3

	e^2	*e^{2}*	e^2

Category	*Numerical*	*LaTeX Equivalent*	*Rendered Form*
	x^(1/2)	*x^{\frac{1}{2}}*	$x^{\frac{1}{2}}$
	sqrt(2)	*\sqrt{2}*	$\sqrt{2}$
	sqrt[3]{8}	*\sqrt[3]{8}*	$\sqrt[3]{8}$
Mathematical Constants	*pi*	*\pi*	π
	e	*e*	e
	i	*i*	i
Golden Ratio	*phi*	*\phi*	ϕ
Trigonometry	*sin(pi/2)*	*\sin(\pi/2)*	$\sin(\pi/2)$
	cos(0)	*\cos(0)*	$\cos(0)$
	tan(π/4)	*\tan(\pi/4)*	$\tan(\pi/4)$
Logarithms	*ln(2)*	*\ln(2)*	$\ln(2)$

	log10(100)	\log_{10}(100)	$\log_{10}(100)$
	log2(x)	\log_{2}(x)	$\log_{2}(x)$
Set Notation	{1, 2, 3}	\{1, 2, 3\}	$\{1, 2, 3\}$
	{}	$\emptyset$	$\emptyset$
	R=(−∞,∞)	$\mathbb{R}$	$\mathbb{R}$
	p,q∈Z (integers)	$\mathbb{Q}$	$\mathbb{Q}$

Category	Numerical	LaTeX Equivalent	Rendered Form
	N={1,2,3,4, …}	$\mathbb{N}$	$\mathbb{N}$
	{x| x>0}	$\{ x \mid x > 0 \}$	$\{ x \mid x > 0 \}$
	x ∈A	x \in A	$x \in A$
	x ∉A	x \notin A	$x \notin A$
	A ∪B	A \cup B	$A \cup B$
	A ∩ B	A \cap B	$A \cap B$
	A \ B	A \setminus B	$A \setminus B$
	P(A)	\mathcal{P}(A)	$\mathcal{P}(A)$
	∀x ∈A	\forall x \in A	$\forall x \in A$

	∃x ∈ A	\exists x \in A	$\exists x \in A$
Combinatorics	C(5, 2)	\binom{5}{2}	$5 2$
	P(5, 2)	{}_5P_2	$\binom{5}{} P_2$
	5!	5!	5!
Integrals	∫_0^1 f(x) dx	\int_{0}^{1} f(x) \, dx	$\int_0^1 f(x)\,dx$
	∫ f(x) dx	\int f(x) \, dx	$\int f(x)\,dx$
	∬_R f(x, y) dx dy	\iint_{R} f(x, y) \, dx \, dy	$\iint_R f(x,y)\,dx\,dy$
	∭f(x, y, z) dx dy dz	\iiint f(x, y, z) \, dx \, dy \, dz	$\iiint f(x,y,z)\,dx\,dy\,dz$

Category	*Numerical*	*LaTeX Equivalent*	*Rendered Form*
	∮_C f(z) dz	\oint_{C} f(z) \, dz	$\oint_C f(z)\,dz$
Matrices	[[a, b], [c, d]]	\begin{bmatrix} a & b \\ c & d \end{bmatrix}	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
	[[a, b, c], [d, e, f], [g, h, i]]	\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}	$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$
	I	\mathbf{I}	$\mathbf{I}$
	det(A)	\det(A)	$\det(A)$

	A^T	A^{T}	A^T
Closed-Form Equations	(-b ± sqrt(b^2 - 4ac))/(2a)	\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}	$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
	(x + y)^n = sum_{k=0}^n C(n,k) x^{n-k} y^k	(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k	$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$
	P(t) = P_0 e^(rt)	P(t) = P_0 e^{rt}	$P(t) = P_0 e^{rt}$
	S = a/(1 - r)	S = \frac{a}{1 - r}	$S = \frac{a}{1-r}$
Approximations (to be specified in prompts if used)	sqrt(2) ≈ 1.414	\sqrt{2} \approx 1.414	$\sqrt{2} \approx 1.414$
	pi ≈ 3.1416	\pi \approx 3.1416	$\pi \approx 3.1416$
	ln(2) ≈ 0.693	\ln(2) \approx 0.693	$\ln(2) \approx 0.693$
Category	***Numerical***	***LaTeX Equivalent***	***Rendered Form***
Vectors	[1, 2, 3]	begin{bmatrix} 1 & 2 & 3 \end	$\begin{bmatrix} 1 & 2 & 3 \end$
	[1 \ 2 \ 3]	\begin{bmatrix} 1 \\ 2 \\ 3 \end	$\begin{bmatrix} 1 \\ 2 \\ 3 \end$

Coordinates	*(3, 4)*	*\[(3, 4) \]*	$(3, 4)$
	(x, y, z)	*\[(x, y, z) \]*	(x, y, z)
Intervals	*(0, 1)*	*\[(0, 1) \]*	$(0, 1)$
	[0,1]	*\[[0, 1] \]*	$[0, 1]$
	[0,1)	*[0, 1)*	$[0, 1)$

Code of Conduct

Expert Code of Conduct

⚠️**PLEASE READ**: It is expected that **all** experts adhere to Project Riga's Code of Conduct. If an expert is found to be within violation of **any** of the following reasons, this could result in immediate dismissal from the project and/or any future collaboration with Snorkel:

Spam:

Your submissions reflect a pattern of short, repetitive, or templated responses that don't meet our standards for original and thoughtful engagement. High-volume, low-value content that disrupts peer workflows and degrades overall task quality.

Plagiarism:

Some of your submissions closely resemble external content or other experts' work. This includes copied prompts/responses, published materials, or reworded internal content. All work must be original and authentic.

Repeated Rejections:

*Work that has **repeatedly** failed peer review due to quality issues, despite being provided support and corrective actions by the Riga Team. While occasional feedback is normal, a consistent pattern of quality issues affects downstream performance.*

Low AHT:

Completion times which suggest that you are prioritizing speed over quality. We encourage you to take your time and be thorough. We expect careful, detailed work that meets our standards.

Frustrated Expert:

Messages to the Snorkel Team or fellow ECs that signal increasing frustration or negativity, via any communication channel. To preserve group cohesion and safety, we expect ECs to maintain respectful and professional relations with Snorkel staff and other ECs at all times.

Insidious Expert:

Interactions that suggest attempts to recruit, solicit, or encourage users to engage with external platforms or agendas. This behavior violates our community guidelines and must cease immediately.

Inappropriate Expert:

Communication or behavior that is seen as being inappropriate, disrespectful, or unprofessional. This includes profane and obscene language, or threats against Snorkel staff and/or other ECs.